Supplementary Material for SoK: Agent Communication Side Channels

The supplement starts with the executed-cover state control, the live TLS-record study and the record-only analysis without call boundaries. Source inspection, coding, the reporting forms and deployment maps follow. The forwarding/concurrency pilot remains a separate final section with its non-rejecting primary tests. Earlier receipt-level timing analyses are preserved in the preserved historical source archive accompanying the LaTeX package.

S1. LLM-planned executed cover

The experiment measures traffic recognition, requested-task completion and full business-state integrity on the same executions. It uses six synthetic ticket actions and six policies: no cover, proxy discard, read cover, hidden isolated execution, live writes and visible isolated execution. All policies share fixtures and frozen random cover draws. The earlier reference-program replay remains a separate artifact; its fixed-position and one-to-four-decoy sweeps are not LLM results.

S1.1. Population, planner and cover semantics

Each of two fixed-weight models, Qwen3:8B and Llama3.1:8B, runs 300 fixtures per condition: 120 training, 60 validation and 120 test, balanced across six requested actions. This gives 3,600 episodes, including 1,440 test executions of 120 unique paired test fixtures. Request phrasings recur across splits; this is fixture holdout, not unseen-template transfer. The seed pairs fixtures and decoy schedules, and condition order rotates. Two workers collect each model cohort on one host.

LangGraph [56] alternates model decisions with actual tool calls. The controller fetches the ticket; required checks constrain action admissibility without naming the correct action or object. The model chooses check order, requested business operation, order identifier, recovery and finish, up to ten decision steps. Temperature is zero, context 4,096 and maximum generated decision length 96 tokens; the fixture supplies the model seed. Fixed manifests and weight blobs match before and after collection.

The fixture contains three orders, customer address, a compensation quota, escalation status and a ledger. Its intended reference is the initial state after exactly the requested legitimate action. Decoys deliberately include operations capable of collateral changes. A frozen schedule places one or two cover calls after the ticket or early honest calls; extra later planner calls do not add cover. Each cover cell triggers all 182 planned test decoys.

Proxy discard computes the response on a disposable copy, retaining no decoy state. Read cover queries state without changing it. Hidden isolated execution applies the decoy on a persistent clone; live write applies it to the primary fixture. Visible isolated execution uses the same clone and schedule, forwarding clearly marked gateway-cover requests and responses to the planner. These objects are not fabricated planner tool calls. In the other conditions, cover responses are hidden from the planner.

S1.2. Transport, observer and endpoints

Seven local MCP Streamable HTTP destinations use actual TLS 1.3/HTTP/2 sockets and a common `/mcp` path. Request bodies use a 4,096-byte padding quantum and responses 16,384 bytes. All seven sessions initialize before each episode and are reused within it. Total TLS cost includes those initializations; call-byte cost and attack features exclude them. The observer receives destination and bidirectional ciphertext byte counts in supplied episode and logical-call windows. It receives no plaintext, model traffic, physical timing, arm label, fixture identifier or private stream assignment. The original replay's variable HTTP-path artifact therefore does not carry over.

Recognition targets the requested action, retaining wrong or failed executions. Fixed features encode the first sixteen call destinations and ciphertext byte counts, per-destination totals and total calls/bytes. LR uses training-only standardization, $C = 1$ and a 3,000-iteration cap. RF uses 300 trees with fixed seed. Validation balanced accuracy selects LR or RF within each model/condition, breaking ties toward LR. Both candidates, the no-cover-selected frozen attack and last/penultimate destination rules remain reported (Table S1). The schedule was frozen before formal collection. The analysis freeze at 17:05 UTC on 10 November 2026 included both position rules with zero terminal test episodes. A logged pre-test amendment at 17:11 handled missing penultimate calls using the training-majority class; it did not change the primary attacker.

Goal completion requires the requested action/object, satisfied prerequisites and the target postcondition. State integrity compares the full business state, including non-target orders, shared fields and the ledger, against the single intended-action reference. The predicates differ: a reference-matching state can still fail the stricter action/object-history

goal predicate. Conversely, a completed goal can leave collateral changes. Table S2 reports both and their completed-but-incorrect-state intersection.

S1.3. Paired inference and uncertainty

The analysis fixes five cover-minus-none contrasts plus visible-minus-hidden for each model. Each endpoint—recognition, completion and integrity—has its own twelve-comparison Holm family. Two-sided exact discordant-pair tests use paired fixture outcomes. Intervals use 10,000 paired fixture bootstrap draws stratified by requested action and condition on the fitted attackers and designed workload. They do not represent template, fit-selection or deployment uncertainty. Table S3 retains every prespecified comparison, including null contrasts.

All ten primary recognition contrasts versus no cover survive correction; the largest adjusted p is 3.08×10^{-8}. Live-write integrity reductions survive for both models (3.70×10^{-32} and 2.98×10^{-19}), while completion survives only for Qwen (.02197, versus 1 for Llama). Every primary visible-minus-hidden contrast has adjusted $p = 1$. These tests do not establish equivalence among cover policies or a privacy upper bound. In particular, Qwen's last-destination rule exceeds the validation-selected learner under cover.

S1.4. Planner paths, costs and limitations

Hidden and visible isolated runs have identical cover responses in all 120 paired fixtures per model. Their honest action sequences differ in 16 Qwen and 65 Llama pairs. Hidden-arm comparisons also vary: proxy versus hidden-isolated differs in one Qwen and fifteen Llama pairs; read versus hidden-isolated differs in zero and fifteen. Pairing fixes the input fixture and schedule, not the realized planner trajectory. These path differences limit mechanism attribution; no repeated-condition cohort estimates the planner noise floor. Mean prompt tokens rise from 1,668.34 to 1,969.49 for Qwen and from 2,251.48 to 2,675.95 for Llama when responses are visible, increases of 18.1% and 18.9%.

Cover call-byte ratios are 1.280–1.346; total TLS ratios are 1.131–1.149 because session initialization is charged to both conditions. Mean local wall times are 8.14–9.94 seconds per test episode, including setup, journaling and two-worker contention. These are local resource measurements. Test outputs contain no invalid decision JSON. Llama reaches the step limit once in no-cover, proxy, read and hidden-isolated conditions, twice in live-write and zero times in visible-isolated; Qwen never does. Every terminal case remains included.

All 3,600 formal episodes, 16,601 model request/response chains, 21,163 tool calls and 4,550 cover calls pass separate state, request and transport checks. Formal unresolved requests are zero. One earlier development request remains unknown, was not retried and is excluded from the formal cohort. The release supplies synthetic planner records, fixture states, lengths, predictions and replay scripts; it excludes credentials, model weights, private TLS keys and raw packet captures.

The adverse writes make the consequence measurable rather than estimating its production prevalence. The experiment is neither a reproduction of WRA's research decoys nor a measurement of physical-timing, model-link or WAN privacy. Llama's unprotected completion and state baselines are already imperfect. The case applies established state-aware evaluation to an adverse cover control, with traffic and cost recorded on those same executions. The direction of collateral harm is strongly predisposed by the chosen writes; the measured endpoint disagreement shows which failures completion omits. Visible-response averages remain inconclusive.

S1.5. Worked application of the evaluation procedure

This application fills the main paper's four-step procedure using saved observations. It adds no data, attacker selection or acceptance threshold.

1. Target and observations. The private target is the requested action among six balanced classes. The observer receives tool destinations and encrypted call-byte counts, with episode and call windows supplied. The model service and physical timestamps are outside the tested view. This original view has no model-link evaluation. The same executions receive a separate client-receipt-timing analysis in the preserved historical source archive; it retains supplied call windows.

2. Allowed execution effects. The synthetic application allows exactly the requested legitimate business action. The reference records its effect on the target and other orders, address, quota, escalation state and ledger. Goal completion additionally requires the correct action/object history and prerequisites. Exact state equality is appropriate to this isolated fixture with no legitimate concurrent updates. A contract can be derived by listing the requested operation's permitted changes, adding frame conditions for all other objects, and specifying quota and ledger updates allowed by that operation. Here the reference executor implements that rule once and the checker compares every field. In production, the contract must also permit legitimate concurrent updates and specify resource budgets; exact fixture equality alone would be inappropriate.

3. Paired measurements. The six conditions share fixtures and cover draws. Every test execution contributes to recognition, completion and state outcomes. LR/RF train on each condition; frozen and destination-position rules are also reported. The joint count of completion with an incorrect state is measured from the two predicates, rather than generally calculated by subtraction. Qwen's live-write arm has 102 completions, of which 95 fail the state contract; Llama has 76, of which 73 fail it. All 36 prespecified paired endpoint comparisons remain in Table S3 and its continuations. Tables S1–S2 give the condition-level outcomes.

TABLE S1. LLM-PLANNED EXECUTED COVER ON 120 PAIRED HELD-OUT FIXTURES PER CELL. THE PRIMARY LR/RF ATTACKER IS SELECTED ON VALIDATION DATA. ALL ATTACK COLUMNS COUNT CORRECT PREDICTIONS OUT OF 120; CHANCE IS 20/120. THE INTERVAL CONDITIONS ON THE FITTED ATTACKER AND SYNTHETIC FIXTURES.

Model	Cover	Frozen	LR	RF	Last	Penultimate	Selected BA [95% CI]
Qwen3:8B	No cover	118	118	117	116	14	0.983 [0.958, 1.000] (LR)
Qwen3:8B	Proxy discard	44	63	62	77	43	0.517 [0.425, 0.608] (RF)
Qwen3:8B	Read cover	42	59	62	77	42	0.517 [0.425, 0.608] (RF)
Qwen3:8B	Isolated, hidden	43	62	61	77	42	0.508 [0.417, 0.600] (RF)
Qwen3:8B	Live write	42	60	62	75	44	0.517 [0.425, 0.608] (RF)
Qwen3:8B	Isolated, visible	40	63	63	77	42	0.525 [0.433, 0.617] (RF)
Llama3.1:8B	No cover	113	113	110	79	64	0.942 [0.900, 0.975] (LR)
Llama3.1:8B	Proxy discard	54	81	79	67	61	0.658 [0.575, 0.733] (RF)
Llama3.1:8B	Read cover	57	78	72	67	62	0.600 [0.517, 0.683] (RF)
Llama3.1:8B	Isolated, hidden	56	81	82	67	60	0.683 [0.600, 0.758] (RF)
Llama3.1:8B	Live write	58	80	75	65	59	0.667 [0.583, 0.750] (LR)
Llama3.1:8B	Isolated, visible	58	79	81	68	64	0.675 [0.592, 0.750] (RF)

TABLE S2. TASK OUTCOMES AND MEASURED COSTS ON THE SAME 120 HELD-OUT FIXTURES. STATE INTEGRITY COMPARES THE ENTIRE BUSINESS STATE WITH THE SINGLE INTENDED-ACTION REFERENCE. BYTE RATIOS ARE MEANS OF PAIRED RATIOS; TOTAL TLS BYTES INCLUDE ALL SEVEN MCP INITIALIZATIONS, WHILE CALL BYTES EXCLUDE INITIALIZATION.

Model	Cover	Goal	State	Goal + state error	Call bytes ×	Total bytes ×	Model calls
Qwen3:8B	No cover	114/120	116/120	0/120	1.00	1.00	4.06
Qwen3:8B	Proxy discard	114/120	116/120	0/120	1.34	1.15	4.06
Qwen3:8B	Read cover	114/120	115/120	0/120	1.34	1.15	4.07
Qwen3:8B	Isolated, hidden	114/120	115/120	0/120	1.34	1.15	4.07
Qwen3:8B	Live write	102/120	7/120	95/120	1.34	1.15	4.08
Qwen3:8B	Isolated, visible	113/120	114/120	1/120	1.35	1.15	4.08
Llama3.1:8B	No cover	82/120	69/120	13/120	1.00	1.00	5.23
Llama3.1:8B	Proxy discard	80/120	68/120	13/120	1.29	1.14	5.31
Llama3.1:8B	Read cover	84/120	71/120	14/120	1.28	1.13	5.25
Llama3.1:8B	Isolated, hidden	83/120	70/120	13/120	1.28	1.13	5.25
Llama3.1:8B	Live write	76/120	3/120	73/120	1.29	1.14	5.35
Llama3.1:8B	Isolated, visible	83/120	73/120	11/120	1.29	1.14	5.33

4. Interpretation against the contract. Live writes fail the declared state requirement even when the target operation completes. Qwen's hidden isolated arm gives a descriptive inference–cost point: the original timing-free attack maximum is 77/120 versus 118/120 without cover, completion is 114/120 in both, integrity is 115/120 versus 116/120, and call bytes are 1.34 times baseline. This maximum summarizes the reported attacks; it does not replace the validation-selected primary attacker or its corrected tests. The added endpoint-timing removal attack instead reaches 112/120 for this isolated condition. No application acceptance tolerances or privacy upper bound were established; the original point cannot be used as its protection level against the expanded observer. The same record exposes what a deployment assessment still needs: broader observers, workload transfer, state tolerances and stronger schedule-aware attacks.

TABLE S3. GOAL COMPLETION: EVERY PRESPECIFIED PAIRED CONTRAST. DIFFERENCES AND CONDITIONAL INTERVALS ARE PERCENTAGE POINTS. THIS ENDPOINT HAS ONE TWELVE-COMPARISON HOLM FAMILY.

Model	Condition minus reference	Difference	Conditional 95% CI	Raw p	Holm p
Qwen3:8B	Proxy – No cover	0.00	[0.00, 0.00]	1.000	1.000
Qwen3:8B	Read – No cover	0.00	[0.00, 0.00]	1.000	1.000
Qwen3:8B	Hidden clone – No cover	0.00	[0.00, 0.00]	1.000	1.000
Qwen3:8B	Live write – No cover	-10.00	[-15.00, -5.00]	0.002	0.022
Qwen3:8B	Visible clone – No cover	-0.83	[-4.17, 2.50]	1.000	1.000
Qwen3:8B	Visible clone – Hidden clone	-0.83	[-4.17, 2.50]	1.000	1.000
Llama3.1:8B	Proxy – No cover	-1.67	[-5.00, 1.67]	0.625	1.000
Llama3.1:8B	Read – No cover	1.67	[-2.50, 5.83]	0.688	1.000
Llama3.1:8B	Hidden clone – No cover	0.83	[0.00, 2.50]	1.000	1.000
Llama3.1:8B	Live write – No cover	-5.00	[-10.83, 0.83]	0.210	1.000
Llama3.1:8B	Visible clone – No cover	0.83	[-6.67, 8.33]	1.000	1.000
Llama3.1:8B	Visible clone – Hidden clone	0.00	[-7.50, 7.50]	1.000	1.000

TABLE S4. REQUESTED-ACTION RECOGNITION: EVERY PRESPECIFIED PAIRED CONTRAST. DIFFERENCES AND CONDITIONAL INTERVALS ARE PERCENTAGE POINTS. THIS ENDPOINT HAS ONE TWELVE-COMPARISON HOLM FAMILY.

Model	Condition minus reference	Difference	Conditional 95% CI	Raw p	Holm p
Qwen3:8B	Proxy – No cover	-46.67	[-55.83, -37.50]	2.78×10^{-17}	3.05×10^{-16}
Qwen3:8B	Read – No cover	-46.67	[-55.00, -37.50]	2.78×10^{-17}	3.05×10^{-16}
Qwen3:8B	Hidden clone – No cover	-47.50	[-56.67, -38.33]	1.39×10^{-17}	1.67×10^{-16}
Qwen3:8B	Live write – No cover	-46.67	[-55.83, -37.50]	2.78×10^{-17}	3.05×10^{-16}
Qwen3:8B	Visible clone – No cover	-45.83	[-55.00, -36.67]	8.05×10^{-16}	6.44×10^{-15}
Qwen3:8B	Visible clone – Hidden clone	1.67	[-5.00, 8.33]	0.815	1.000
Llama3.1:8B	Proxy – No cover	-28.33	[-35.83, -20.83]	1.08×10^{-9}	5.59×10^{-9}
Llama3.1:8B	Read – No cover	-34.17	[-42.50, -25.83]	9.09×10^{-13}	6.37×10^{-12}
Llama3.1:8B	Hidden clone – No cover	-25.83	[-34.17, -18.33]	9.31×10^{-10}	5.59×10^{-9}
Llama3.1:8B	Live write – No cover	-27.50	[-35.83, -19.17]	1.02×10^{-8}	3.07×10^{-8}
Llama3.1:8B	Visible clone – No cover	-26.67	[-35.00, -19.17]	4.07×10^{-9}	1.63×10^{-8}
Llama3.1:8B	Visible clone – Hidden clone	-0.83	[-8.33, 6.67]	1.000	1.000

TABLE S5. BUSINESS-STATE INTEGRITY: EVERY PRESPECIFIED PAIRED CONTRAST. DIFFERENCES AND CONDITIONAL INTERVALS ARE PERCENTAGE POINTS. THIS ENDPOINT HAS ONE TWELVE-COMPARISON HOLM FAMILY.

Model	Condition minus reference	Difference	Conditional 95% CI	Raw p	Holm p
Qwen3:8B	Proxy – No cover	0.00	[0.00, 0.00]	1.000	1.000
Qwen3:8B	Read – No cover	-0.83	[-2.50, 0.00]	1.000	1.000
Qwen3:8B	Hidden clone – No cover	-0.83	[-2.50, 0.00]	1.000	1.000
Qwen3:8B	Live write – No cover	-90.83	[-95.00, -85.83]	3.08×10^{-33}	3.70×10^{-32}
Qwen3:8B	Visible clone – No cover	-1.67	[-5.83, 1.67]	0.688	1.000
Qwen3:8B	Visible clone – Hidden clone	-0.83	[-4.17, 2.50]	1.000	1.000
Llama3.1:8B	Proxy – No cover	-0.83	[-4.17, 2.50]	1.000	1.000
Llama3.1:8B	Read – No cover	1.67	[-2.50, 6.67]	0.727	1.000
Llama3.1:8B	Hidden clone – No cover	0.83	[-1.67, 3.33]	1.000	1.000
Llama3.1:8B	Live write – No cover	-55.00	[-62.50, -47.50]	2.71×10^{-20}	2.98×10^{-19}
Llama3.1:8B	Visible clone – No cover	3.33	[-4.17, 10.83]	0.523	1.000
Llama3.1:8B	Visible clone – Hidden clone	2.50	[-5.00, 10.00]	0.664	1.000

S2. Live TLS-record timing and planner-generated cover

This study tests cover on the network records produced by actual dispatch. It compares immediate controller cover, sequential timing matching, concurrent dispatch and a second planner that generates a complete cover execution. The source literature, the earlier receipt-level counterfactuals and this collection retain separate denominators.

S2.1. Population, protocol and execution

The study reuses the 300 executed-cover fixtures per model, with 120 training, 60 validation and 120 test fixtures, balanced over six requested actions. Qwen3:8B and Llama3.1:8B each execute all five arms, giving 3,000 formal episodes and 120 paired held-out fixtures per cell. Phrasings recur across splits. This is a new timing-layer collection on previously used workloads; earlier cover results were known when the protocol was designed.

LangGraph runs the same business planner against seven local MCP Streamable HTTP services over TLS 1.3/HTTP/2. Model weights and templates are fixed by layer hashes before and after collection. Model options are temperature zero, a 4,096-token context and at most 96 generated tokens per model step. One episode is active at a time; the local inference server has two execution slots so that the shadow chain can run alongside the real chain. Connections are shared by the two chains within an episode. A lock serializes calls to the same service, while calls to different services and planner inference may overlap. The experiment consequently exercises real concurrent chains without claiming unrestricted same-service multiplexing.

Every arm uses a common `/mcp` request path and 4,096/16,384-byte request/response body-padding quanta. The real planner receives only its own tool responses. Cover actions execute on isolated state and cannot directly write the real fixture. All requests and terminal responses are journaled; state outcomes are recomputed by replaying the synthetic operations. No episode is replaced after a missing response.

Five arms. *No cover* supplies the baseline. *Immediate* dispatches the original one or two cover calls directly after their triggering honest response. *Matched, sequential* waits for a sampled gap and holds the server response for a sampled minimum duration, blocking the real chain as required by that schedule. *Matched, concurrent* uses the same draws in a persistent episode-scoped dispatcher while the real planner continues inference. *Planner shadow* runs a complete second episode of the same model on an independently generated ticket, including its own checks, actions and feedback. It shares the service connections but not business state or planner context.

The shadow action is sampled uniformly from all six classes, including the real class. An independent fair draw selects which chain starts first; the later chain begins 0.2–1.5 s after the first ticket response. The attack is not told which first call is real. Random order removes that particular role cue, while the chain-separation attack tests whether other role information remains.

Historical timing pools. All no-cover episodes are collected first. Only their 120 training fixtures per model supply honest inter-call gaps and per-destination durations; the pools and collection amendments are frozen and pushed before any formal cover run. Timing-matched arms sample from those pools. A monotonic not-before wait enforces the selected gap, and the server holds a cover response until its selected minimum duration has elapsed. Transport, padding, scheduling and concurrent work still contribute to the achieved record times. The cover-arm order rotates across fixtures. Backend prompt caches persist between arms; a reused real ticket and a new shadow ticket can therefore have different cache histories.

S2.2. Observation and independent timing check

The primary observer receives destinations, bidirectional ciphertext counts, episode/call windows and relay timestamps at the completion of each encrypted TLS record. It receives no message body, operation, task identifier, ground-truth call role or model-link observation at test inference. Initialization precedes the call windows. TLS 1.3 outer type 23 identifies the encrypted records used to delimit a window; it is not interpreted as an application-plaintext type.

Npcap independently records both ingress directions. TCP sequence reassembly, separated by connection generation, must reproduce every relay record's outer type, length and count. The evaluator then measures relay-versus-packet completion-time error. Relay read completion is a software clock, not an exact NIC arrival timestamp. Supplied call boundaries and destination visibility remain observer privileges; the experiment does not evaluate discovery of those boundaries.

S2.3. Attacks, selection and prespecified decisions

Size-only candidates are standardized logistic regression, a 300-tree random forest, the frozen no-cover classifier, and last- and penultimate-destination rules. Timing candidates add direct timing-plus-size classification, supervised gap-detector removal, label-free two-component Gaussian-mixture gap removal, context-detector removal and joint decoding of zero, one or two removals. The context detector includes destinations, positions, sizes and within-episode timing relations. LR/RF choices and detector thresholds are selected on validation data; detector selection uses call-role F1 and then precision. Call-role annotations are permitted for training and evaluation, never as test-time inputs.

The shadow-specific attack partitions the observed calls into two causal chains with a beam of 64 candidates and a transition-gap model fitted on no-cover training traffic. A clean-chain classifier predicts each chain's action. A role selector, fitted on labelled shadow training episodes, chooses the real chain using observed chain summaries. It is not given the test roles. Exact partition recovery, real-chain

recovery and call-role accuracy are descriptive diagnostics. With perfect recovery of an unordered pair of independent uniform six-class labels, guessing the real label succeeds with probability $7/12$; this is an idealized pair-ambiguity reference, not a general privacy bound.

For each arm and model, validation accuracy selects a size-only attack and an adaptive attack from the complete suite; ties favor the declared size-first order. Test maxima are reported separately as descriptive results and do not replace the frozen choices. The test unit is the paired fixture, with 120 outcomes per cell.

Four testing families. Family P contains the six adaptive-minus-size comparisons for the sequential, concurrent and shadow arms. Each uses exact paired McNemar testing with Holm correction across the six comparisons. An arm meets the prespecified separability rule if either model gains at least 12/120 with adjusted $p < .05$. Family E tests the same six contrasts with paired t-TOST, margin $\pm 12/120$, 119 degrees of freedom and Holm correction. Both models must pass for an arm to meet the equivalence rule. At all ties, the implementation uses the conservative boundary probability $.9^{120}$ instead of zero. Equivalence concerns the validation-selected adaptive attack relative to the selected size attack; it does not establish cover-role indistinguishability or an all-attacker guarantee.

Family R has two immediate-arm replication comparisons: label-free removal against the selected size attack. Replication requires a gain of at least 24/120 and Holm-adjusted $p < .05$ in both models. Family C has eight size-only cover-versus-no-cover comparisons, also exact paired tests with Holm correction. C quantifies the size-view change; remaining adaptive inference is reported alongside it. Ten thousand action-stratified paired fixture bootstrap draws give intervals conditional on the fitted attacks and the reused workload. They omit refitting, attack-selection and workload-selection uncertainty. With 120 pairs, moderate increments or equivalence can remain unresolved.

S2.4. Costs and business outcomes

Real-chain latency begins at admission before either shadow chain starts and ends when the real planner finishes, including shadow-first waiting but excluding session initialization. Unique whole-connection bytes include setup and teardown without double-counting overlapping windows. Model calls and tokens include both planners in the shadow arm. Requested-goal success, whole-state correctness and honest-path agreement with no cover are reported on the same fixtures. Changes in planner behavior, cache history and shared inference resources accompany these policy comparisons, so an endpoint difference alone cannot isolate one causal mechanism.

S2.5. Complete attack and decision results

Tables S6–S7 give every frozen candidate; Table S8 gives every test, discordance count and conditional interval. Immediate removal meets R in both models: 112 versus 75 and 108 versus 80, with adjusted $p = 1.9 \times 10^{-8}$ and 4.3×10^{-6}. For historical sequential dispatch, P rejects for Llama (104 versus 82; $p = .00078$) but not Qwen (93 versus 76; $p = .0559$). Concurrent dispatch rejects for both (99 versus 66 and 106 versus 68). Each history-based arm consequently meets its prespecified separability rule through at least one model. The context attack uses non-timing inputs as well as time; these comparisons do not isolate the causal contribution of timestamps.

Both shadow selectors choose the same size attack for the size and adaptive roles, so their predictions are identical. Family E therefore passes by construction ($p_{\mathrm{Holm}} = 1.94 \times 10^{-5}$ under the all-tie safeguard). We classify this cell as degenerate: it is reported for completeness and is not counted as evidence that trace distributions or real/cover roles are indistinguishable. Section S2.6 gives the comparison that replaces it. Qwen's last-destination rule wins validation at 35/60, ahead of RF at 33/60, but on test scores 54/120 against RF's 70/120. Llama's selected RF scores 64/120. These selection differences are kept visible instead of choosing a new primary attack after test evaluation. All eight size-view cover-versus-none comparisons in C reject; the adaptive results show why a size-view decrease alone is an incomplete protection assessment.

Shadow roles and start order. The chain attack exactly partitions both chains in 48/120 Qwen and 59/120 Llama fixtures, but exactly selects the real chain in only 29 and 32. Its role assignments are correct on 563/977 and 673/1,255 calls; these calls are dependent within fixtures. Test data contain 53 shadow-first and 67 real-first fixtures per model, with 21 matching real/shadow action labels. The RF scores 28/53 and 42/67 for Qwen, and 25/53 and 39/67 for Llama, respectively. These descriptive strata expose order sensitivity without adding an inferential family. Shadow tasks themselves reach 116/120 and 89/120 requested goals. The fixed beam, role learner, recurring phrasings and different cache histories limit the tested attack suite.

S2.6. Pair-ambiguity floor for planner shadows

Let the real chain and k shadow chains carry labels $L_0, \ldots, L_k$ from C classes, with L_0 the real label. An idealized attacker observes the unordered multiset $\{L_0, \ldots, L_k\}$, which corresponds to perfect chain separation and per-chain classification, but not the index of the real chain. Because the start order is randomized and the shadow ticket is generated by the same process, the real chain is exchangeable with the shadows given the multiset. Hence $\Pr[L_0 = c \mid \text{multiset}] = n_c/(k+1)$, where n_c counts chains labeled c, and the Bayes-optimal guess is a modal label. Its success probability is

$$P^*(k) = \frac{\mathbb{E}[\max_c n_c]}{k+1}.$$

For iid uniform shadow labels and $C = 6$, exact enumeration gives 1, $7/12$, $53/108$, $197/432$ and $185/432$ for $k = 0, \ldots, 4$. If shadow labels are instead drawn without replacement from the $C - 1$ classes other than the real one, every

TABLE S6. ALL FROZEN QWEN3:8B ATTACKS ON LIVE-RECORD TEST OBSERVATIONS: CORRECT OUT OF 120 PER CELL. NONE IS NO COVER; HISTORY-BASED POLICIES SAMPLE MINIMUM GAP/RESPONSE HOLDS. DASHES INDICATE AN ATTACK NOT APPLIED TO NO-COVER TRAFFIC. VALIDATION CHOICES ARE REPORTED SEPARATELY FROM THESE INDIVIDUAL TEST SCORES.

Attack	None	Immediate	History seq.	History concurrent	Shadow
Size/destination LR	117	64	61	52	60
Size/destination RF	116	59	61	52	70
Frozen no-cover classifier	117	40	41	40	31
Last destination	114	75	76	66	54
Penultimate destination	14	43	42	47	43
Direct timing + size	104	60	57	47	64
Supervised gap removal	–	112	67	57	20
Label-free mixture removal	–	112	53	38	34
Context removal	–	111	93	87	20
Joint removal/task decoding	–	112	99	99	32
Two-chain separation	–	–	–	–	49

TABLE S7. ALL FROZEN LLAMA3.1:8B ATTACKS ON LIVE-RECORD TEST OBSERVATIONS: CORRECT OUT OF 120 PER CELL. NONE IS NO COVER; HISTORY-BASED POLICIES SAMPLE MINIMUM GAP/RESPONSE HOLDS. DASHES INDICATE AN ATTACK NOT APPLIED TO NO-COVER TRAFFIC. VALIDATION CHOICES ARE REPORTED SEPARATELY FROM THESE INDIVIDUAL TEST SCORES.

Attack	None	Immediate	History seq.	History concurrent	Shadow
Size/destination LR	110	82	79	68	59
Size/destination RF	112	80	82	68	64
Frozen no-cover classifier	110	54	57	57	43
Last destination	80	65	65	48	43
Penultimate destination	66	65	64	64	46
Direct timing + size	110	77	81	75	59
Supervised gap removal	–	108	100	98	20
Label-free mixture removal	–	108	14	66	43
Context removal	–	108	101	103	20
Joint removal/task decoding	–	108	104	106	33
Two-chain separation	–	–	–	–	34

multiset has distinct labels, the posterior is uniform over the $k{+}1$ chains, and $P^*(k) = 1/(k{+}1)$ for $k \le C{-}1$. This variant keeps each shadow's marginal label uniform. The script `experiment/shadow_floor.py` computes these values and writes `analysis/shadow-floor.json`.

On a realized split with m matching pairs among n fixtures and $k = 1$, the floor is $(m{+}(n{-}m)/2)/n$. The live test split has $m = 21$ and $n = 120$, giving 70.5/120. The strongest tested call-window attack scores 70/120 (Qwen) and 64/120 (Llama), and the validation-selected attacks score 54 and 64, so none exceeds the realized floor. The floor is an idealized reference rather than a bound: a real attacker can exceed it only by extracting role information, so the role-relevant comparison uses the 99 fixtures whose labels differ, where the floor is exactly one half. Aggregate counts restrict the forest's correct predictions on those fixtures only to $[49, 70]$ and $[43, 64]$. The exact two-sided binomial test against one half requires the saved per-fixture predictions; these predictions are part of the artifact.

S2.7. Achieved timing and the independent clock

Table S9 distinguishes sampled holds from achieved record durations. Each row includes all 455 cover calls in that arm/model, including training and validation. Sequential Qwen cover averages a .5198 s sampled gap and .5660 s achieved anchor gap; a .0405 s hold becomes a .1004 s observed duration. Concurrent duration increases to .1712 s. Llama's .0768 s mean hold becomes .1097 s sequential and .1234 s concurrent. Thus the live policy does not instantiate the exact timing-distribution transform used by the receipt-level counterfactual. Its measured inference, overhead and altered chronological context must be evaluated directly.

All 143,920 no-cover and 629,530 cover records match independently reassembled TCP/TLS records, across 21,000 connections and 19,244 call windows. All-record median absolute completion-time errors are 1.07 and 1.33 ms, with 99th percentiles 16.38 and 21.68 ms; the maximum is 1.675 s, including setup and teardown. Table S10 restricts the comparison to call endpoints: the maximum absolute error is 148.39 ms, and no episode's call order changes under the packet clock.

The separately labelled packet-clock diagnostic refits only the label-free mixture/removal calculation. Its task scores exactly reproduce the relay-clock scores in all eight cover cells, including 112 and 108 for immediate cover. On those held-out immediate calls it flags 181/182 and 182/182 decoys, with zero false flags among 490 and 634 honest calls. This is a sensitivity check for removal on supplied call windows; P/E remain relay-clock primary analyses, and the

TABLE S8. ALL PRESPECIFIED LIVE-RECORD TEST FAMILIES. P/E COMPARE THE VALIDATION-SELECTED ADAPTIVE AND SIZE ATTACKS; R COMPARES LABEL-FREE IMMEDIATE REMOVAL WITH SIZE; C COMPARES EACH COVER SIZE ATTACK WITH NO COVER. +/− COUNT FIXTURE DISCORDANCES. INTERVALS ARE 95% PAIRED ACTION-STRATIFIED BOOTSTRAP INTERVALS CONDITIONAL ON THE FITTED ATTACKS. E USES PAIRED T-TOST WITH A ±10-POINT MARGIN; THE SHADOW SELECTORS CHOOSE THE IDENTICAL SIZE ATTACK, SO ALL PAIRS TIE. HOLM CORRECTION IS SEPARATE WITHIN EACH DECLARED FAMILY.

Family	Model	Arm	+/−	Δ (pp)	Conditional interval	Raw p	Holm p
P	Qwen	History seq.	32/15	14.2	[3.3, 25.0]	0.0186	0.0559
P	Qwen	History concurrent	42/9	27.5	[17.5, 38.3]	3.4×10^{-6}	1.7×10^{-5}
P	Qwen	Shadow	0/0	0.0	[0.0, 0.0]	1	1
P	Llama	History seq.	28/6	18.3	[10.0, 26.7]	2×10^{-4}	7.8×10^{-4}
P	Llama	History concurrent	41/3	31.7	[22.5, 40.8]	1.6×10^{-9}	9.7×10^{-9}
P	Llama	Shadow	0/0	0.0	[0.0, 0.0]	1	1
E	Qwen	History seq.	32/15	14.2	[3.3, 25.0]	0.7713	1
E	Qwen	History concurrent	42/9	27.5	[17.5, 38.3]	0.9992	1
E	Qwen	Shadow	0/0	0.0	[0.0, 0.0]	3.2×10^{-6}	1.9×10^{-5}
E	Llama	History seq.	28/6	18.3	[10.0, 26.7]	0.9643	1
E	Llama	History concurrent	41/3	31.7	[22.5, 40.8]	1	1
E	Llama	Shadow	0/0	0.0	[0.0, 0.0]	3.2×10^{-6}	1.9×10^{-5}
R	Qwen	Immediate	41/4	30.8	[21.7, 40.0]	9.3×10^{-9}	1.9×10^{-8}
R	Llama	Immediate	33/5	23.3	[15.0, 32.5]	4.3×10^{-6}	4.3×10^{-6}
C	Qwen	Immediate	0/42	-35.0	[-43.3, -26.7]	4.5×10^{-13}	2.7×10^{-12}
C	Qwen	History seq.	0/41	-34.2	[-42.5, -25.8]	9.1×10^{-13}	4.5×10^{-12}
C	Qwen	History concurrent	0/51	-42.5	[-50.8, -34.2]	8.9×10^{-16}	6.2×10^{-15}
C	Qwen	Shadow	0/63	-52.5	[-60.8, -43.3]	2.2×10^{-19}	1.7×10^{-18}
C	Llama	Immediate	3/33	-25.0	[-33.3, -16.7]	2.3×10^{-7}	4.5×10^{-7}
C	Llama	History seq.	2/30	-23.3	[-31.7, -15.0]	2.5×10^{-7}	4.5×10^{-7}
C	Llama	History concurrent	2/44	-35.0	[-44.2, -25.8]	3.1×10^{-11}	1.2×10^{-10}
C	Llama	Shadow	4/50	-38.3	[-47.5, -28.3]	3.8×10^{-11}	1.2×10^{-10}

study makes no physical-network confidentiality claim.

S2.8. Resource costs, paths and business outcomes

Table S11 reports both the primary real-chain latency and a descriptive all-work duration through all planner chains and tool-response receipts. This prevents a concurrent dispatcher from hiding queued work behind real-planner completion. Mean all-work duration is 3.369 and 2.835 s for concurrent history dispatch, and 5.543 and 4.643 s for shadow execution. The corresponding no-cover means are 2.912 and 3.332 s. These are warm-session costs from this machine and execution order, not deployment throughput or a causal speedup estimate.

Whole-connection byte factors are 1.13–1.15 for controller policies and 1.31–1.35 for shadows; restricting accounting to supplied call windows gives 1.26–1.33 and 1.68–1.71. Shadow model-token factors are 2.01 and 1.99. Goal/state outcomes for shadow are 112/113 and 82/67 of 120, against no-cover 113/114 and 82/70. Honest paths exactly match no cover on 115/120 and 103/120 shadow fixtures, so even hidden isolated cover does not guarantee identical planning. No business-equivalence hypothesis is substituted for the prespecified accuracy equivalence test.

The formal dataset contains 3,000 primary episodes plus 600 shadow chains; every issued request has a terminal record and no unknown outcome was replaced. Active collector wall time is 38.8 min for no cover and 193.1 min for the four cover arms, including their service initialization and model loads but excluding the intervening pool/audit gate and later analyses. The payload-free release reconstructs all primary features, predictions, test arithmetic and synthetic-state outcomes. It excludes model/tool message bodies, raw captures, keys and weights; private capture reconstruction and its errors have separate receipts.

TABLE S9. ACHIEVED HISTORY-BASED COVER TIMING, ALL SPLITS (455 COVER CALLS PER ROW). MEANS IN SECONDS. AN ANCHOR GAP IS MEASURED FROM THE ACTUAL PRIOR COMPLETED ANCHOR CALL TO COVER RECORD START. SAMPLED DURATIONS ARE MINIMUM SERVER HOLDS, WHILE ACHIEVED DURATIONS INCLUDE RECORD-LEVEL TRANSPORT AND SCHEDULING.

Model	Arm	Drawn gap	Achieved anchor gap	Drawn hold	Achieved duration
Qwen	History seq.	0.5198	0.5660	0.0405	0.1004
Qwen	History concurrent	0.5198	0.5772	0.0405	0.1712
Llama	History seq.	0.5865	0.6203	0.0768	0.1097
Llama	History concurrent	0.5865	0.6204	0.0768	0.1234

TABLE S10. CALL-WINDOW CLOCK DIAGNOSTICS ACROSS ALL 300 FIXTURES PER CELL. RELAY VERSUS INDEPENDENT PACKET COMPLETION ERRORS, IN MILLISECONDS. VALUES ARE ABSOLUTE-ERROR MEDIAN / 99TH PERCENTILE / MAXIMUM. EVERY CALL RETAINS ITS CHRONOLOGICAL ORDER UNDER THE PACKET CLOCK. SETUP/TEARDOWN RECORDS ARE EXCLUDED FROM THIS TABLE.

Model	Arm	Call windows	Start: median / p99 / max	End: median / p99 / max
Qwen	None	1218	0.58 / 7.67 / 18.01	0.42 / 5.89 / 9.21
Qwen	Immediate	1673	1.70 / 8.57 / 32.76	1.23 / 4.20 / 7.98
Qwen	History seq.	1671	1.46 / 8.03 / 10.62	1.07 / 4.36 / 7.02
Qwen	History concurrent	1672	1.74 / 26.06 / 42.48	1.36 / 11.96 / 89.31
Qwen	Shadow	2411	2.17 / 57.62 / 148.39	1.72 / 25.46 / 116.16
Llama	None	1536	0.84 / 7.55 / 68.03	0.62 / 5.27 / 8.95
Llama	Immediate	1998	0.58 / 4.49 / 17.04	0.41 / 2.31 / 5.63
Llama	History seq.	1996	0.50 / 4.19 / 9.63	0.39 / 2.40 / 4.50
Llama	History concurrent	2002	0.58 / 5.54 / 71.35	0.43 / 6.65 / 29.04
Llama	Shadow	3067	0.72 / 12.10 / 99.22	0.64 / 9.71 / 89.08

TABLE S11. LIVE TEST-SET COSTS. REAL AND ALL ARE MEAN SECONDS FROM ADMISSION TO REAL-PLANNER FINISH AND TO ALL PLANNER/TOOL-RESPONSE COMPLETION, RESPECTIVELY; BOTH EXCLUDE SESSION SETUP/TEARDOWN. RECORD SPAN IS THE MEAN FIRST-TO-LAST TOOL-RECORD OBSERVATION. UNIQUE/CALL BYTES ARE SUMMED-BYTE FACTORS VERSUS NO COVER FOR COMPLETE CONNECTIONS AND SUPPLIED CALL WINDOWS. CALLS COUNT ALL TOOL/MODEL CALLS PER FIXTURE; TOKEN FACTORS INCLUDE THE SHADOW PLANNER. THESE ARE OBSERVED COSTS UNDER PERSISTENT CACHES AND PHASE ORDER, NOT ISOLATED CAUSAL SPEED MEASUREMENTS.

Model	Arm	Real s	All s	Span s	Unique×	Call×	Tool calls	Model calls	Tokens×
Qwen	None	2.912	2.912	2.272	1.00	1.00	4.08	4.08	1.00
Qwen	Immediate	3.575	3.575	2.899	1.15	1.33	5.60	4.08	1.00
Qwen	History seq.	4.280	4.280	3.611	1.15	1.32	5.59	4.08	1.00
Qwen	History concurrent	3.328	3.369	2.896	1.15	1.33	5.60	4.08	1.00
Qwen	Shadow	4.804	5.543	4.681	1.31	1.68	8.14	8.14	2.01
Llama	None	3.332	3.332	2.759	1.00	1.00	5.24	5.24	1.00
Llama	Immediate	2.916	2.916	2.420	1.13	1.27	6.80	5.28	1.01
Llama	History seq.	3.822	3.822	3.339	1.13	1.26	6.80	5.28	1.01
Llama	History concurrent	2.791	2.835	2.448	1.13	1.27	6.83	5.31	1.02
Llama	Shadow	3.939	4.643	4.087	1.35	1.71	10.46	10.45	1.99

S3. Record-only analysis without supplied call endpoints

This extension reuses all 3,000 main episodes. The analyst fixed its code and candidate grid before computing new test scores, with the older task outcomes already known. It is retrospective reused-workload evidence, not a new cohort. The underlying live collection, primary choices and tests remain unchanged.

S3.1. Inputs and observation scope

Each episode contributes seven final, closed connection logs, linked by the episode's connection bundle. All outer-type23 records are included; record time is made relative to the earliest such record. Initialization and closing records are retained. The feature whitelist is relative time, destination, direction and ciphertext record length. Labels and split identifiers join evaluation rows but never enter features. No per-call terminal, operation, private role or HTTP stream mapping is read by the attacker path. A source-hash manifest binds every input. Whole-task association is retained, so this is removal of call endpoints, not task localization.

S3.2. Candidate representations and fitting

Within each destination connection, records separated by more than 5, 20, 50, 100 or 250 ms form separate proposed events. Events retain the first and last timestamps and directional ciphertext sums. Destination summaries include event count, directional bytes, record count and maximum event volume; timing variants also include first/last times and total event duration. Ordered features use the first four and last sixteen events, with destination one-hot and log byte counts, adding duration and preceding gap in timing variants. Raw-record candidates treat each record as an event and use the same summaries. No plaintext or true call interval is required.

LR uses standardization, $C = 1$ and at most 3,000 iterations. RF has 300 trees. The fixed seed is 111121 and numerical libraries use two threads. Per-condition training fits each representation. A two-component mixture fits log gaps from that condition's training events without role labels. The crossing between its short/long components defines removal; the first event is retained as a heuristic. The remaining events go to a no-cover-trained task LR/RF at the same segmentation/view. Validation chooses baseline, removal and overall attacks by task accuracy with fixed-order ties. The selection file is written before that cell's test predictions.

There are forty-four fixed-grid candidates per model/condition: five gaps, two views, two learners for refitted baseline and clean-model removal, plus four raw-record candidates. Each cell retains 120 training, 60 validation and 120 test fixtures. The primary comparisons are selected removal minus selected baseline on immediate cover, with exact paired McNemar and Holm across the two models. Ten thousand paired test-fixture bootstrap draws give conditional intervals; they omit model selection, refitting and workload-selection uncertainty. Other arms and test maxima are descriptive. A test maximum is neither an independently selected confirmatory attack nor an upper bound on all possible inference.

S3.3. Exploratory protocol-aware eventization

After the Qwen idle-grid test results were known, a separate plan fixed one public-protocol alternative. An upward encrypted record of at least 4,096 bytes starts a proposed transaction. The first two such requests per connection correspond to the public MCP initialize/initialized setup sequence and are removed from this representation. A transaction ends at the last downward record of at least 4,096 bytes

before the next trigger, or at its start if no such response exists. This rule exploits the configured padding quantum and same-service serialization; it reads no true operation, call endpoint or private role. It is not an HTTP/2 plaintext parser or a general multiplexing solution.

The same size/timing views, LR/RF and clean-model GMM removal yield eight further candidates per cell. TRAIN/validation/test remain fixed, and validation selections are saved before their test predictions. These are exploratory results with no new superiority family. The first fixed-grid results and tests are preserved in full. The displayed Max spans all fifty-two candidates, while B/R still refer to the original fixed-grid validation choices. Complete candidate counts and predictions are in the local analysis package.

S3.4. Complete results

The independent boundary diagnostic is computed only after attack predictions, using saved private call truth. It is not an attacker feature or a segmentation selection criterion. The local package retains every candidate validation score, held-out prediction, input hash and reconstruction script. Individual record metadata and predictions remain local.

TABLE S12. SELECTED RECORD-ONLY ATTACKS. B/R/A REFER TO VALIDATION-SELECTED FIXED-GRID BASELINE, REMOVAL AND OVERALL ATTACKERS. PROTO B/R GIVES THE SEPARATE EXPLORATORY PROTOCOL-AWARE SELECTION; MAX SPANS BOTH SUITES. COUNTS ARE OUT OF 120. COMPLETE CANDIDATE OUTCOMES REMAIN IN THE LOCAL ANALYSIS PACKAGE.

Planner	Policy	B	R	Proto B/R	Max	Selected fixed-grid attacks (B; R; A)
Qwen	No cover	115	109	117/93	117	events 5ms LR; remove 20ms timing RF; events 5ms LR
Qwen	Immediate	80	69	68/91	91	records timing RF; remove 5ms RF; records timing RF
Qwen	History, sequential	73	74	66/45	75	records timing RF; remove 5ms RF; remove 5ms RF
Qwen	History, concurrent	56	61	51/47	62	events 5ms LR; remove 5ms RF; remove 5ms RF
Qwen	Planner shadow	69	65	66/52	73	events 20ms RF; remove 100ms timing RF; events 20ms RF
Llama	No cover	108	101	109/88	112	events 5ms timing LR; remove 20ms RF; events 5ms timing LR
Llama	Immediate	90	86	90/87	97	records RF; remove 5ms timing RF; records RF
Llama	History, sequential	88	56	90/71	93	events 5ms timing RF; remove 20ms LR; events 5ms timing RF
Llama	History, concurrent	78	62	83/68	88	events 50ms RF; remove 5ms LR; events 50ms RF
Llama	Planner shadow	69	56	68/52	69	events 5ms RF; remove 5ms RF; events 5ms RF

TABLE S13. PRESPECIFIED RECORD-ONLY IMMEDIATE-COVER COMPARISONS. DISCORDANCE COUNTS ARE REMOVAL-ONLY/BASELINE-ONLY CORRECT. INTERVALS CONDITION ON FITTED PREDICTORS AND FIXTURES.

Planner	Removal	Baseline	$+/-$	Difference pp	Conditional 95% interval	Raw p	Holm p
Qwen	69	80	3/14	-9.2	[-15.8, -2.5]	0.01273	0.02545
Llama	86	90	10/14	-3.3	[-11.7, 4.2]	0.5413	0.5413

TABLE S14. INDEPENDENT CALL-BOUNDARY AUDIT, IOU .5, AFTER PREDICTION. EACH ROW USES120 ASSOCIATED TEST EPISODES. IDLE SELECTS ITS BURST GAP BY VALIDATION TASK ACCURACY; PROTOCOL USES PUBLIC PADDING/SETUP RULES. VALUES ARE PRECISION / RECALL / F1, INCLUDING SPURIOUS PROPOSED EVENTS. THESE TRUTH COMPARISONS ARE NOT ATTACKER INPUTS.

Planner	Policy	Idle-gap eventization	Protocol-aware eventization
Qwen	No cover	0.000/0.000/0.000	0.233/0.434/0.303
Qwen	Immediate	0.000/0.000/0.000	0.229/0.412/0.295
Qwen	History sequential	0.000/0.000/0.000	0.272/0.489/0.350
Qwen	History concurrent	0.000/0.001/0.000	0.266/0.478/0.342
Qwen	Shadow	0.012/0.078/0.021	0.390/0.612/0.476
Llama	No cover	0.000/0.000/0.000	0.294/0.510/0.373
Llama	Immediate	0.000/0.000/0.000	0.302/0.512/0.380
Llama	History sequential	0.000/0.000/0.000	0.378/0.641/0.476
Llama	History concurrent	0.357/0.802/0.494	0.358/0.606/0.450
Llama	Shadow	0.000/0.000/0.000	0.428/0.637/0.512

S4. Selection, Observation Contracts and Source Checks

S4.1. Fixed-window coverage protocol

The search protocol was recorded before retrieval on 9 November 2026. Eligibility uses first-submission dates from 1 January 2024 through 9 November 2026 and requires feedback-driven Agent execution linked to communication-side inference or an evaluated protection mechanism. Transit metadata, recipient-visible requests and active probes remain separate observers. Whole semantic trajectories, browser instrumentation alone, model-serving-only targets, deliberate covert encoding and generic deception without an identification endpoint have separate dispositions. The purpose is bounded coverage, not a claim of exhaustive recall.

Twelve fixed public-web topic queries cover Agent traffic, tools, MCP, research and browser/terminal fingerprints. Native requests use arXiv, DBLP and Semantic Scholar endpoints; the latter two returned no usable records because of access/rate limits. They are failed acquisition channels, not zero-result searches. The arXiv query returned all 180 reported records with a limit of 200. Its exact expression is:

```
(all:"LLM agent" OR all:"AI agent" OR
all:"tool call" OR all:MCP OR
all:"web agent" OR all:"research
agent" OR all:agentic) AND (all:"side
channel" OR all:"side-channel" OR
all:"traffic analysis" OR
all:fingerprint OR all:"network
privacy" OR all:"timing attack") AND
submittedDate:[202401010000 TO
202611092359]
```

All executed strings and receipts are in `coverage-2024-2026/`. There were 38 public-web queries including topic, identity and citation follow-ups; these do not constitute 38 independent databases. The twelve topic queries displayed 101 distinct URLs, representing 33 scholarly candidate records after excluding news, tools and unresolved non-paper pages. Deduplication with arXiv gives 209 records. Two additional entered citation leads yield 211 screened records: fourteen direct, forty adjacent, 155 outside and two unresolved paper identities. The retained CHeaT seed gives fifteen included sources over 212 distinct returned/entered/retained records. Search-engine total-hit estimates are not used as denominators. Earlier query results are excluded from this screening denominator.

One-step backward screening inspected 88 reference entries in the Trace, AgentTrace, HTTP-signature and honeypot full texts. Title/citation follow-ups supplied additional identity checks. Trace cites both AgentPrint and the 2024 SSH honeypot; the latter was absent from the native query. Its inspected version is v2 (2025), with first submission in 2024. AgentSnare's inspected abstract describes adaptive deception against known Agent adversaries without a separate communication-identification endpoint; it remains adjacent. The same inclusion rule excludes prompt-injection defenses without a communication-inference endpoint.

Other borderline cases retain explicit access limits. AgentProv's categorical tool-choice probes do not execute a feedback loop; SigLeak uses whole semantic action/observation trajectories, consistent with the existing IntentMiner boundary. LIDAR uses harness instrumentation and repository changes; browser-event fingerprints require DOM/UI access. None is counted as an encrypted-transit replication. The full screening sheets preserve titles, versions, primary URLs and reasons, including unresolved identities.

S4.2. Core identities and actual inspection

Table S15 fixes all fifteen inspected sources. WRA and CHeaT are verified publications and their selected final-text passages were rechecked. The HTTP work's two-page ACM workshop record is verified by publisher-deposited DOI metadata (10.1145/3843282.3844445); its longer v3 supplies the detailed extraction here. Trace's arXiv API metadata names WAITI 2026, but no matching official paper entry was verified and the December event is after the cutoff. Twelve sources, including Trace, have no additional independently verified publication identity. This is an audit result, not proof of non-publication.

Check coverage. The inspection inventory covers all fifteen sources, including selected final-PDF checks of WRA and CHeaT. Numerical checks locate displayed results; no new source experiment was replicated and no new source dataset was obtained. Original HTML/PDF hashes remain local, with public URLs and selected-check descriptions in the review package. No independent Web-archive snapshot is claimed. Publication identity, actual check method and conclusion dependency remain separate attributes.

Checks of identification protocols. AgentTrace's 50 training/50 test probe split, stochastic trials, query-budget table and repair ablation were read together. At fifty probes, repeated trials reuse the same held-out probe identities. Trace's within-scaffold, held-out-scaffold and proprietary-scaffold results have different units or populations; clean-run rejection and injected-run counts are explicit. The HTTP study provides descriptive request signatures and three trials per service; neither an identification accuracy nor a held-out split is supplied. The honeypot's eight candidates are detector-defined, with unknown field precision and recall. These boundaries remain attached to the respective rows.

S4.3. Setting matrix and interpretive checks

The `field-map/evidence-matrix.csv` records forty-four settings. Its companion target sheet records label basis, execution outcome, failure handling and policy context. The main maps cover all fifteen sources, including descriptive mechanisms and controls as such. Numeric values are not normalized across reconstruction, classification, candidate detection and prevention endpoints.

Composition and scaffold transfer. Communication graphs holds out one of three compositions within each of nine known labels, with 270 workflows. Its .18 result

TABLE S15. VERSION-LOCKED DIRECT-SOURCE IDENTITIES AND EXTRACTION LOCATORS. PRIMARY LINKS ARE IN THE BIBLIOGRAPHY AND REVIEW MATRIX.

Source	Inspected version	Locations
AgentPrint [1]	2510.07176v1	App. B–D
WRA [2]	USENIX Security 2026 final	Secs. 3–4, 6, 8; Tables 8–11
Communication graphs [3]	2606.07150v3	Secs. 9.1, 9.3, 9.7–9.8, 9.12–9.13
Cloud benchmark [4]	2609.31877v2	Secs. III–V
PersonaFingerprint [5]	2605.15962v1	Secs. 2, 3.3, 5
MARK [6]	2606.20910v1	Secs. III, VI; Table IV
Fayolle et al. [7]	2606.30119v1	Secs. 3, 5–8
AgentTell [8]	2609.32915v1	Secs. 3.2, 4.3, 5; Apps. B.4, D.3
ANT [9]	2610.06514v1	Secs. 3–4; Table 3; model-view figure
CHeaT [10]	USENIX Security 2025 final	Secs. 5.2, 6.2, 7; Tables 3–4
PrivacyTrace [35]	2610.41885v1	Secs. 3–5; Fig. 6
AgentTrace [47]	2610.32092v1	Secs. 2–4; Tables 1–4; Fig. 3
Trace [48]	2605.01186v1	Secs. 2–4
HTTP signatures [49]	2604.02544v3 extension	Secs. 3.3–3.6; Tables 1–3
LLM Agent Honeypot [51]	2410.13919v2	Secs. 3–5

is not unknown-label recognition. Trace instead holds out one of three Agent scaffolds for model-family attribution, reporting macro-F1 .815 versus .981 in the mixed-session test. The separate proprietary-scaffold 78% accuracy uses a narrower model set. These are named transfer axes, not interchangeable estimates of generalization.

Live adaptation and costs. The inspected `scripts/eval_ohttp_live.py` path passes live captures through `evaluate_binding` in `binding.py`, `evaluate_views` in `evaluate.py` and fold-wise training in `cross_validate` in `classify.py`. This establishes protected-capture RF refitting. A separate stronger-search output covers direct/Tor/Nym/shim conditions and is not assigned to the live OHTTP row. The six-class .30 live result and its approximately thirteen-fold bytes/eleven-fold latency cannot be paired with the nine-class .67 workflow score as a before/after comparison.

WRA conditions. The official final PDF reports five-decoy functional recovery .565 versus .770 without defense in Table 8 (query reconstruction); Table 9 concerns trait reconstruction. Visibility Table 11 instead pairs 100%/10% visibility with recovery .770/.610 and report utility 8.55/8.54. The latter utility does not price executed decoys. Earlier selected-code inspection of `config_defence.json` and `utils_ICL_prompt.py` locates decoy mixing on test traces, while ICL examples call `sample_traces` with ratio 1.0. The full training-folder contents were not exhaustively inspected; this selected path does not establish the absence of every other adaptation configuration. Matched protected examples remain unresolved. These final-text checks cover the stated passages.

ANT and recipient objectives. ANT's approximate absolute model-view accuracies are reconstructed from the displayed full scores and rounded changes: .737, .450, .679, .665, .713, .689, .594 and .637. Risk counts use a separate 209-episode population, while primitive windows use runtime boundaries. CHeaT separates request-trigger presence checks, password-output controls and CTF prevention. Its adaptive payload-detection FPR is not the FPR of a natural-traffic Agent classifier. The main maps keep these denominators and objectives separate.

Links to the mechanism synthesis. The seven factual comparison groups retain their original definitions. Execution organization uses the named-transfer and failure-signal groups; distributed exposure uses observation, label, policy and model-view statements; cover costs use the matched-intervention statements. Section S8 separates their observed basis from design implications and proposed tests. The `synthesis/claims.csv` protocol joins and `mechanism-synthesis/` interpretation links are complementary. Neither verifier establishes semantic source correctness.

S4.4. Unified premise withdrawal

The analysis follows thirty-three source statements in seven comparisons across twenty scenarios, yielding 140 retained/withdrawn rows. Each premise has a source, setting and locator. Scenarios remove each of fifteen sources, retain matched inspected publications only, remove either known shared-author pair, or withhold the live-refitting code premise. The pairs are AgentPrint/ANT and WRA/MARK; shared authors motivate concentration checks, not assertions of shared data. The statements cover label definitions, selection, security references, observation units, transfer, containment and failure handling.

J1 requires an objective and an action example; J2 requires success-selection and a contrasting first-pass/null/refusal rule; J3 requires the risk comparator, disclosure-rule example and an outcome check. The failure-signal contrast requires both its Trace selection premise and AgentTrace repair premise. Other comparisons retain individual propositions without treating transfer axes as substitutes. Table S17 summarizes the retained statements and sources; `field-map/source-`

TABLE S16. INSPECTION COVERAGE FOR ALL FIFTEEN DIRECT SOURCES. EACH ROW NAMES SELECTED CHECKS; CODE OR DATA INSPECTION IS STATED ONLY WHERE PERFORMED. NO SOURCE EXPERIMENT WAS INDEPENDENTLY RERUN. VERSION IDENTITIES AND DETAILED LOCATORS ARE IN TABLE S15; THE REVIEW PACKAGE PRESERVES THE RECEIPTS.

Source / text	Text and numerical check	Implementation / data check	Remaining boundary
AgentPrint, v1	Apps. B–D; identity F1 and separate occupation endpoint.	Author-bound training/data not located on inspected surfaces.	Behavior-F1 and pool totals conflict.
WRA, official final	Secs. 3–8; query-decoy Table 8 and visibility Table 11 distinguished.	Selected ICL/configuration paths; training folder not exhaustively inspected.	Persona count conflict; protected examples unresolved.
Communication graphs, v3	Secs. 9.1, 9.3, 9.7–9.8, 9.12–9.13; transfer, live scores and costs.	Four live-evaluation files pinned; protected-data refitting traced.	Live utility and broader attack search remain separate.
Cloud benchmark, v2	Secs. III–V; task/topology scores and selected v1/v2 passages.	Code and episode data not independently inspected.	Capture-total conflict retained.
PersonaFingerprint, v1	Secs. 2, 3.3, 5; session split and site scores.	Code and episode data not independently inspected.	Pooled-summary discrepancy retained.
MARK, v1	Secs. III, VI; Table IV; layer F1 and first-pass selection.	Code and episode data not independently inspected.	Combined browser view includes non-network features.
Fayolle et al., v1	Secs. 3–8; visit, fingerprint and blocking populations.	Code and episode data not independently inspected.	Visit success is a narrow outcome.
AgentTell, v1	Secs. 3.2, 4.3, 5; Apps. B.4, D.3; disclosure, failure and null result.	Code and episode data not independently inspected.	Loaded/cold differs from guardrail-on/off.
ANT, v1	Secs. 3–4; eight model-view scores and thirteen risk confusion matrices reconstructed.	Repository shell accessed; episode/code files not obtained.	Rounded-table arithmetic; no raw-data reproduction.
CHeaT, official final	Secs. 5.2, 6.2, 7; Tables 3–4; detection, prevention and adaptation.	Code and episode data not independently inspected.	Active payload controls differ from passive classification.
PrivacyTrace, v1	Secs. 3–5; Fig. 6; metadata, prior and argument views.	Code and episode data not independently inspected.	Below-prior metadata result retained.
AgentTrace, v1	Secs. 2–4; Tables 1–4; Fig. 3; probe split, budget and repair ablation.	Code and episode data not independently inspected.	Repeated trials reuse held-out probe identities.
Trace, v1	Secs. 2–4; within-, held-out- and proprietary-scaffold results; rejection counts.	Code and episode data not independently inspected.	Transfer rows have different populations and metrics.
HTTP signatures, v3	Secs. 3.3–3.6; Tables 1–3; three trials per service.	Code and episode data not independently inspected.	Descriptive signatures; short workshop identity is separate.
LLM Agent Honeypot, v2	Secs. 3–5; eight detector-defined field candidates.	Code and episode data not independently inspected.	Field precision and recall unknown.

`dependence.json` records every row. Strict publication filtering retains WRA and CHeaT. The HTTP short-paper identity alone does not retain claims extracted only from the extended version.

Five unresolved numerical fields remain outside these synthesis premises: AgentPrint's behavior F1 and capture-pool totals, WRA's persona test count, the cloud benchmark's capture totals and PersonaFingerprint's pooled summaries. Exact conflicting values and locators remain in the input record. No preferred value is silently substituted.

S4.5. Expanded source attack maps

TABLE S17. CONCLUSION-LEVEL SOURCE WITHDRAWAL. PREMISES ARE MAPPED STATEMENTS; THE THREE SOURCE-COUNT COLUMNS COUNT DISTINCT PAPERS RETAINING AT LEAST ONE MAPPED PREMISE. THEY ARE NOT INDEPENDENT CONFIRMATIONS. "MIN. ONE OUT" REMOVES ONE SOURCE AT A TIME. THE PUBLISHED-TEXT FILTER REQUIRES A MATCHED INSPECTED PUBLICATION.

		Distinct sources			
Comparison	Statements	Full	Min. one out	Published text	Dependence on sources
Label objects	4	4	3	1	Objective/action contrast survives one-source removal; removing AgentPrint and ANT together removes the action-label side.
Selected populations	6	6	5	0	Success-selection contrasts with first-pass, null and refusal handling. Trace adds a crash/no-command exclusion.
Target / policy rules	4	4	3	1	The task-risk comparator needs ANT; the disclosure-rule example needs AgentTell.
Observation units	8	7	6	1	Several sources supply units; only the honeypot supplies these field-candidate counts, without external identity labels.
Deployment transfer	3	3	2	0	Composition, topology and scaffold are different axes, each supported by its own source.
Protection across views	6	4	3	1	Six statements come from four papers: communication graphs and WRA supply two each; the cloud benchmark and ANT supply one each.
Session exclusion / step repair	2	2	1	0	Trace excludes crashed/command-free sessions; AgentTrace observes recoverable parser errors. Both sources are needed; the failure events differ.

TABLE S18. ATTACK FIELD MAP, PANEL A: TRANSIT OBSERVATIONS. SCORES RETAIN THEIR METRIC, TASK AND SUPPLIED UNIT; THEY ARE REPRESENTATIVE REPORTED OPERATING POINTS, NOT A CROSS-PAPER RANKING. THE MATRIX RECORDS ALL FORTY-FOUR SETTINGS. P: INSPECTED PREPRINT; V: INSPECTED PUBLISHER VERSION.

Interface / observer	Work / target	Observable features / learner	Evaluation unit	Reported result and boundary
Client–agent; transit	AgentPrint [P] [1]; hosted agent	Directional packet counts/bytes in 1,800 relative-time bins; CNN	Supplied browser interaction	Mixed-flow macro-F1: .8660 closed, .8477 open; 50 monitored agents. Behavior-F1 conflict is kept separate.
Agent–Web; transit	WRA [V] [2]; research query	Contacted domains and timing; ICL or fine-tuned LLM	Supplied research trace	Functional query recovery .770 with domains; .725 with domains plus timing in the compared visibility views. This is a reconstruction score.
Inter-agent / tools; transit	Communication graphs [P] [3]; workflow	Message/graph, size and timing descriptors; random forest	Workflow record	Nine-class accuracy .67; within-label composition holdout .18. The live defense test instead has six classes.
Agent–model; transit	Cloud benchmark [P] [4]; assigned task	247 traffic descriptors; random forest	Logged successful run	Ten-class balanced accuracy .853/.860 for graph/star coordination. Size-excluded feature protocols score .540/.587.
Agent–Web; transit	PersonaFingerprint [P] [5]; browsing persona	Signed lengths/interarrival times; CNN representation and prediction heads	Fixed packet window; session split	Per-site macro-F1 about .78–.91; pooled-summary discrepancy excluded from synthesis. Explicit other-class tests have a different population.
Combined egress, M or remaining links; transit	ANT [P] [9]; task, primitive, risk	Direction/length/time; 13 learners including DF and TikTok	Supplied episode or logged stage	Five-task accuracy .431–.727; 47-primitive accuracy .355–.462. Risk test: 20 positive/189 negative, requiring recall and false alarms.

TABLE S19. ATTACK FIELD MAP, PANEL B: CONTACTED RECIPIENTS AND ACTIVE AUDITORS. THESE OBSERVERS SEE REQUESTS ADDRESSED TO THEM OR ISSUE PROBES; THEIR RESULTS DO NOT MEASURE ENCRYPTED-TRANSIT INFERENCE. E: INSPECTED EXTENDED PREPRINT WITH A SEPARATELY VERIFIED PUBLISHED SHORT VERSION. NR: NO RECOGNITION RATE REPORTED IN THE MAPPED SETTING. ASR: ATTACK SUCCESS RATE.

Interface / access	Work / target	Observable features / learner	Reported result and evaluation boundary
Web recipient	MARK [P] [6]; client identity	TLS/HTTP2 and headers; decision tree	F1 .725 for TLS/HTTP2, .799 for headers; .971 combined with browser instrumentation, a richer view. First-pass trials.
Web recipient	Fayolle et al. [P] [7]; client / access	Protocol signatures; richer behavior classifier and site challenges	Fingerprint analysis excludes failed visits. Access success is separately verified through form/cookie/page identifiers; not a privacy score.
Documentation recipient	HTTP signatures [E] [49]; service/client	Request method/version, headers and origin; descriptive comparison	Nine coding agents and six assistants, three trials each; NR. Request signatures are observations, not a held-out identification rate.
Terminal recipient	Trace [P] [48]; model family	Command TF-IDF unigrams/bigrams; linear SVM	Seven families; 2,028 clean/injected sessions: macro-F1 .981; held-out scaffold .815. A separate proprietary-scaffold test reports 78% accuracy.
Tool interface; active audit	AgentTrace [P] [47]; model identity	Serialized arguments, parser errors and repair order; Markov likelihood	Six models, 50 held-out probes per trial: 96.0% accuracy; 10 probes: 92.8%. Repeated stochastic trials reuse probe identities.
Web recipient; active choice	AgentTell [P] [8]; prior-origin fact	Which offered action is requested; loaded/cold comparison	Matching choice in 61.1% of 7,630 loaded sessions; the leakage measure subtracts cold matching. Actions share a completable probe task.
Tool recipient / gateway	PrivacyTrace [P] [35]; profile attribute	Tool metadata/order; content-added LLM inference	Metadata ASR 14.53%, schema prior 17.97%; arguments 27.71%. 68,000 attribute slots are clustered within 4,000 trajectories.
Contacted asset; active challenge	CHeaT [V] [10]; agent presence	Planted URL following; separate password-output control	Presence-detection mechanisms, NR for natural-traffic classification. Capture-the-flag prevention is a distinct endpoint (Supplement S7).
SSH recipient; active probes	LLM Agent Honeypot [P] [51]; possible agent	Prompt compliance and response-time heuristic	Three-month deployment: eight candidate agents among 8,130,731 interactions. No externally verified field labels or detection precision/recall.

S5. Mechanism Layer and Correspondence

The common inclusion criterion is an original result about an execution-dependent signal, necessary observation access, a detection/transfer condition or a protection action that can be mapped to an Agent setting. The sixteen communication anchors cover interactive Web state leakage, distributed request tracing, IoT/voice event detection, deployment assumptions, residual features and transport cover. Six serving anchors are included by the same criterion: Time Will Tell, Whisper Leak, Early Bird, CachePrune, Weiss et al. and Carlini–Nasr. The latter group includes inspected preprints. All twenty-two have separate locators and transfer limits in `mechanism-sources.csv`. The main transfer matrix groups the selected relations by the condition needed for an Agent application; it does not turn the anchor set into additional Agent replications.

The original communication extension used twelve recorded discovery/bibliographic queries and selected twelve anchors from fourteen entered leads. WAP5 and vPath require host observations; their graph reconstruction is not a passive encrypted-link result. A retrieved serverless data-flow lead was excluded because it tracks sensitive data with a different observation model. The search does not prove the absence of a passive microservice or serverless attack. Serving mechanisms were integrated from the already inspected context sources, not counted as a second independent discovery sample.

Mechanism anchors never enter Agent-source support counts. They establish why the missing capability test is relevant, not its result for an Agent. Transport specifications likewise state what MCP, HTTP/2 and TLS support without demonstrating the privacy of a particular deployment. The artifact records the seven correspondence groups and their setting-level links.

The transfer matrix uses four relationship labels. Retained applies to a classical obligation that remains relevant, such as accounting for unknown boundaries or protected-condition attack adaptation. Extended names an extra execution condition, such as following a tool result into a later model request or tracking the original-task effects of executed cover. Access mismatch rejects a proposed correspondence between host instrumentation and passive packet observation. Untested identifies a performance claim for which the mapped Agent evidence supplies no matched test. A row's relation and the number/publication status of its sources are separate attributes. In particular, STP's tunnel terminator remains valid for Agent deployments that actually use it; WRA's executed research is a different intervention. Classical work also includes multiple observation locations and cross-device dependencies, so a universal single-link assumption is not part of this synthesis.

The focused WF extension adds four published anchors. Wang–Goldberg [45], Sections 3–5, supplies the realistic splitting/background precedent. k-FP [46], Sections 4–5, supplies the open-world/feature reference. BAPM [43] is checked through its official ACSAC abstract and author slides; no quantitative result is imported. ARES [44] is identified by its IEEE publication record and author repository; the inspected extended threat model is arXiv:2501.12622v1, Sections III–IV. That extension is not silently substituted for the 2023 experimental results. These additions refine the localization/transfer correspondence without changing Agent-source support counts.

S6. Labels, Outcomes and Policy Context

The extraction distinguishes the label being predicted from the action recorded, the original task's outcome and its policy context. The forty-four rows in `field-map/target-evidence.csv` retain the source version, setting, passage locator and each field's status. *Reported* means an explicit assessed endpoint for a stated execution population, including a source-defined success rate or a probe/parser outcome. *Partial* records a measured component or restricted population; *not located* denotes an applicable execution outcome not found in the inspected scope. *Not applicable* requires no new execution population, such as preference-only analysis or offline capture transformation. A logged action or absent task-success evaluation alone does not make the field inapplicable. These statuses describe evidence availability, not study quality.

Three contrasts illustrate the rules. WRA's prompt target is an assigned research query, whereas ANT-stage uses log-derived primitive labels; recognizing the first does not establish the second. Trace excludes forty of 1,050 planned clean runs, while AgentTrace uses parser outcomes and ordered repairs as probe signals. These are session eligibility and within-session behavior, respectively, not a paired estimate

TABLE S20. MECHANISM LOCATORS AND THE OBSERVATION RELATIONSHIP USED IN THE SOK.

Source	Locator	Correspondence boundary
Chen et al. [28]	III.B–C	Successive Web-state observations constrain inputs.
Chapman–Evans [29]	Sec. 3	Browser exploration supplies labels and actions.
WAP5 [30]	Secs. 1–3	Participating-host socket traces support path inference.
vPath [31]	Secs. 2.2, 3	Thread/network observations and request–reply assumptions.
PINGPONG [33]	IV.B, V.B–C	Stream signature detection and negative traffic.
Ahmed et al. [22]	Secs. 3.1–3.2, 4, 7	Invocation localization before command recognition.
Juarez et al. [15]	Secs. 3–6	Environment, client and open-world assumptions.
Peek-a-Boo [32]	Sec. I; Fig. 1	Coarse time/volume features remain useful under padding.
Mathews et al. [16]	Sec. I, findings	Adapted attacks and matched costs.
STP [34]	Secs. 6.1, 7.1	Activity/cover shaping; tunnel endpoints drop cover.
Walkie-Talkie; FRONT [17], [18]	Secs. 3–4; Sec. 4	Page-load shaping mechanisms need an Agent workload correspondence.
Time Will Tell [13]	Secs. III, VI	Exact token counts and latency proxies are different views.
Whisper Leak [12]	Secs. 3.1, 6.1–6.3	TLS size/gap inference; new captures and transformed traces differ.
Early Bird [11]	II-C, III, V-A–V-B	Prefix and semantic-cache defenses have distinct costs.
CachePrune [14]	Secs. 3.1, 4, 5.7	Filter-disabled component cost does not price enabled protection.
Weiss et al. [20]	Sec. 2; attack construction	Encrypted streaming token-length sequences.
Carlini–Nasr [21]	Secs. II–III	Optimization-dependent timing; passive/active access separated.

of the effect of failure filtering. ANT-risk labels benchmark misuse; AgentTell evaluates action choice under a disclosure rule, and CHeaT reports a defined CTF-prevention outcome. None supplies a general per-user authorization label for arbitrary executions.

Each contrast remains tied to its sources. The source-withdrawal analysis in Section S4.4 states which required premises survive each removal. The complete extraction is provided as CSV in the review artifact, alongside the full earlier typeset table in `historical-studies/r98-supplement-archive.pdf`. Labels, outcomes and policy fields have not been inferred from classifier accuracy or completed from missing source information.

S6.1. Coding-manual pilot and source adjudication

The released author encoding (AU) is the coding of record: the author decided every code against the cited passage, using AI only to locate passages and draft candidates. Two later AI passes (AI-1, AI-2) re-coded the same forty-four settings to test the codebook, not the coder. The supplied provenance identifies AI-1 and AI-2 as the same model, not independent people. AI-1 had seen aggregate counts; AI-2 had seen AU's status values for ten settings. Table S21 therefore describes a non-blind coding-manual pilot and sensitivity analysis. Its bootstrap intervals resample settings despite source clustering and dependent encodings; they are not human-coder reliability bounds.

Independent human re-coding. The decision-critical subset (44 access classes and protection flags, nine intervention settings, six transfer relations) was released as a blind packet with a codebook. An independent security PhD researcher who had not seen the manuscript or AU coded it from the sources without AI tools, recording a supporting passage for every code. Agreement with AU, before any discussion, is 44/44 for access class ($\kappa = 1$), 9/9 for intervention purpose ($\kappa = 1$), 4/4 each for protected-condition adaptation and same-intervention utility in the privacy settings ($\kappa = 1$), 38/44 for the protection flag ($\kappa = .61$) and 5/6 for transfer relations ($\kappa = .77$); overall 104/111 decisions. The privacy denominator is identical. Four protection-flag disagreements mark with/without contrasts that AU did not code as protection comparisons: the composition section of the communication-graph study, its active front-running setting, and the two web-agent fingerprinting settings with stealth configurations. Two mark AgentTrace settings that the coder read as lacking an unspoofed baseline. The transfer disagreement is T1 (retained versus extended). None of the seven can create a query/workflow privacy setting with both an adapted attacker and matched utility, because neither the communication-graph study nor the identity-obfuscation and blocking settings report original-task utility. The author then adjudicated each disagreement against the cited passages. The coder's code was adopted in six cases: CG-composition (its Sec. 9.12 ladder re-measures the CG-transform properties and is counted with that setting), both Fayolle fingerprinting settings (identity obfuscation), ATR-main and ATR-budget (no unspoofed baseline), and T1 (extended). AU's code was retained for CG-active, whose with/without contrast concerns an integrity endpoint rather than inference. After adjudication, three settings are identity obfuscation, thirty-four have no protection comparison, and the four-setting privacy denominator and the matched-intervention synthesis are unchanged. The coder's sheets, the agreement output and the adjudication record are in the review artifact.

Removing the ten AI-2-exposed settings leaves thirty-four rows: AU–AI-2 κ is .785 for failure handling and .433

TABLE S21. ORIGINAL, UNADJUDICATED CODING PILOT ON FORTY-FOUR SETTINGS. PAIRWISE COLUMNS ARE COHEN'S κ; THE THREE-PASS COLUMN IS FLEISS' κ WITH A 2,000-DRAW SETTING-BOOTSTRAP INTERVAL. FULL AGREEMENT IN THIS PILOT DOES NOT ESTABLISH FIELD VALIDITY.

Field	AU–AI-1	AU–AI-2	AI-1–AI-2	Three-pass κ [95%]	All agree
Access, protection, named transfer, open world (each)	1.00	1.00	1.00	1.00	44/44
Failure handling	.46	.80	.63	.63 [.46, .75]	28/44
Execution outcome	.15	.46	.52	.37 [.23, .48]	14/44

for outcomes. This does not remove the shared-model or non-blind design. The number coded *reported* also depends on applicability. Failure handling is 19/42, 18/43 and 16/43 for AU, AI-1 and AI-2; outcomes are 13/38, 7/38 and 15/43. These are descriptive coding ranges with different denominators, not a common-denominator confidence interval. Neither the original 13/38 nor an AI majority is treated as a settled field-coverage estimate.

Rules that resolve the ambiguity. First name the execution and criterion. A source-defined success/failure count qualifies as *reported*, with the strength of its criterion recorded separately. Explicit parser/repair outcomes qualify for a probe study, not as original-task utility. Persona-alignment diagnostics assess a behavioral component and are *partial*. Raw action labels or final messages alone do not establish an evaluated task outcome. Reserve *not applicable* for a setting with no new execution population; missing success evidence in an execution study is *not located*.

Fifteen disputed records. Source inspection changes thirteen outcome codes and retains two, with a per-setting ledger. All four ANT settings become *not located*: the original codes were two partial, one reported and one inapplicable, not four inapplicable. Four cloud-benchmark settings become *reported* because Section IV-A states 4,070 successful and 42 failed runs (98.9%). This is an inherited collection-level criterion, not four independent success measurements; the source's conflicting capture totals remain unresolved. Three PersonaFingerprint settings become *partial* for consistency/diversity diagnostics. AgentPrint behavior becomes *partial* for capability activation/retention, and HTTP signatures becomes *not located* for an evaluated execution outcome.

The communication-graph active setting remains *reported*: Section 9.5/Table 4 supplies live front-running outcomes over 100 rounds per condition, which both AI passes missed. Its offline capture-transformation setting remains *not applicable* because the transformation does not re-execute tasks. These source checks take precedence over votes. The revised sheet contains 15 reported outcomes among 42 applicable rows; twenty-nine rows retain their prior extraction rather than receiving a new independent coding. No post-adjudication κ is presented as validation.

The artifact preserves all original encodings, exact agreement calculations, the exposed-row sensitivity, the clarified manual, source versions and the fifteen-row before/after ledger. General execution-outcome reporting remains distinct from original-task utility under the *same defense*; consequently these revisions do not alter the four-setting privacy/utility denominator in the main text.

S7. Evidence Availability and Reporting Examples

S7.1. Eligibility and evidence-availability rules

The descriptive tally uses the existing forty-four setting records from fifteen direct sources. It adds no retrieval or independent coding. The artifact's `transfer-synthesis/setting-availability.csv` carries each record's original access description, intervention, adaptation and utility evidence, source version and locator together with the new grouping. `availability-summary.json` gives the rules and exact membership behind every count.

Access is classified by the rights used in the evaluated setting: transit metadata (22 settings, 6 sources), addressed-recipient observations (10, 5), active intervention/probing (11, 6), or a downstream preference analysis (1, 1). These classes partition settings; source groups overlap. AgentPrint's persona analysis uses an inferred preference representation and is kept separate from a new packet-observer test. CHeaT's planted-token tests use the defender's active setup and knowledge. Combining access classes would grant an observer information from a different experiment.

The privacy denominator comprises `WRA-decoy`, `WRA-visibility`, `CG-transform` and `CG-live`: four query/workflow protection settings from two sources. Protected-condition fitting or an adaptive comparison is located for the two communication-graph settings. This records the inspected adaptation, not an optimal attacker. An original-task quality/goal measure for the same intervention is located for WRA visibility. Extra queries, attacker budgets, bytes and latency alone do not satisfy that field. Hence adaptation is located in 2/4, original-task utility in 1/4, and both in 0/4. Within the sole executed-cover setting, original-task utility is unlocated (0/1). WRA's utility does not transfer from the visibility condition to decoy execution.

Three identity-obfuscation records (AgentTrace's obfuscation setting and the two Fayolle fingerprinting settings with stealth configurations), two adversarial-Agent blocking records (Fayolle and CHeaT), and thirty-four records without a protection comparison are outside that denominator. The communication-graph composition setting re-measures the CG-transform properties on the wire-faithful corpus view and is counted with CG-transform. They remain in the corpus. "Not located" applies to retained evidence, while

"outside denominator" states a different question; neither is a negative result. No field-completion fraction is used to score studies or validate the checklist. The table below states the four included decisions explicitly.

S7.2. Checklist and completed examples

Mark a box when its applicable fields and missing measurements are recorded.

- ☐ **C1. Observer and links.** Name the observed interfaces, available metadata/content, access level and any supplied task boundaries.
- ☐ **C2. Target and ground truth.** State whether labels denote intent, action, success, identity or risk; identify the policy reference for any authorization claim.
- ☐ **C3. Population and failures.** Give denominators and the treatment of crashes, refusals, retries and incomplete executions; report the relevant execution outcome.
- ☐ **C4. Split and transfer.** Name the split unit and each changed deployment axis; distinguish a supplied execution from an event found in a stream.
- ☐ **C5. Attacks and adaptation.** State features, learners and still-visible links; compare unchanged attacks with attacks trained on protected executions.
- ☐ **C6. Matched protection and cost.** Bind residual inference, original-task utility, added calls, bytes and delay to the same intervention and execution population.
- ☐ **C7. Metrics and uncertainty.** Specify metric, denominator and uncertainty assumptions; for monitoring, report conditional error rates and the intended prevalence.
- ☐ **C8. Source and availability.** Identify the source version and result locator, accessible artifacts and the scope of any code or data checks.

These forms apply the Section 8 checklist to three selected settings already in the evidence matrix: WRA's five-decoy condition, the communication-graph live OHTTP condition and ANT's risk test. The authors filled them from the recorded source passages and code checks. They introduce no new source experiment or independent analyst evaluation.

Reading a filled form. *Reported* identifies supplied evidence; *partial* identifies a supplied component and its specific gap. *Not located* applies to the inspected passages or mapped record, not all possible source artifacts. *Not applicable* means that the field is outside this setting's claim. A checkmark in a new study's blank checklist means the field has been filled or explicitly dispositioned; it is not a pass mark. The examples distinguish an unavailable interval in the current extraction from a claim that the source never reports uncertainty.

The completed examples retain different open fields. WRA needs matched decoy adaptation and utility before its recovery reduction supports a complete privacy–utility point. The live OHTTP example already supports fixed-family protected-data refitting and transmission cost, while leaving task utility unresolved. ANT supports a benchmark-risk classifier comparison; an authorization judgment would additionally need the relevant execution policy and principal. None of these distinctions requires changing a reported score.

S8. Mechanism Synthesis and Open-Question Specifications

The main synthesis combines observations already recorded in the forty-four-setting matrix. A source result, a deduction about the protection scope, and a proposed mechanism test have different roles. Table S25 gives the links. The review package's `mechanism-synthesis/` files also connect each conditional-risk row and open question to setting IDs. Those links make the interpretation inspectable; they add no experimental observation.

S8.1. Q1: feature stability under execution changes

Hold the private label, observer rights and measurement layer fixed before comparing an execution change. Split by the changed composition or implementation, and keep repeated runs from the same task or session together. Frozen transfer and local adaptation answer different questions and should use separate results. Retain failures and refusals as measured outcomes, with any eligibility restrictions declared. A joint holdout tests interactions that separate single-axis tests cannot settle. Analysis should identify which feature family changes its predictive utility; an accuracy loss alone does not identify memorization as its cause.

S8.2. Q2: feedback attribution

Use the same initial request for every category and disclose category-dependent information only in the later tool response. Retain prefix measurements as a negative control, predeclare the primary attacker/comparison and minimum effect of interest, and separate task failures from missing responses. Section S10 implements this design with paired full and projected forwarding. The projection removes supplementary records while retaining the task fields; its effect combines changes in content, encoded length and subsequent execution. Separating these mediators requires additional interventions. Where a forced call path would invalidate the task, measure that path as a mediator rather than calling the comparison content-only. The artifact archive retains the earlier common-prefix population and its inconclusive result.

S8.3. Q3: cover termination and semantic isolation

Define the original task's required final state, allowable additional work and resource budget before evaluating cover. A transport-terminated cover event and an executed decoy are different interventions. Isolation should be checked for business-state writes, quota consumption and feedback

TABLE S22. SETTING-LEVEL BASIS FOR THE QUERY/WORKFLOW PRIVACY DENOMINATOR. ORIGINAL RECORD VERSIONS ARE RETAINED; THE OFFICIAL WRA TEXT SUPPLIES THE UPDATED TABLE LOCATOR.

Setting	Adaptation evidence	Original-task utility evidence	Source locator and boundary
WRA decoys	Matched protected ICL examples/refitting unresolved.	Not located for the decoy condition.	Official USENIX text, Sec. 8.1, Table 8; selected test-side mixing code. Five extra research traces do not supply a task-quality result.
WRA visibility	Protected-condition refitting not located.	Located: report utility for the visibility intervention.	Sec. 8.2, Tables 10–11. Not a full live rerouting cost evaluation.
Communication graphs, transformations	Located: protected fitting and adaptive comparison.	Not located.	2606.07150v3, Secs. 9.3, 9.8 and 9.12 (composition corpus). Transformed captures do not measure a live task outcome.
Communication graphs, live OHTTP	Located: fixed-family RF refit per protected capture.	Not located; bytes and latency are transmission costs.	2606.07150v3, Sec. 9.7 and selected live-evaluation code. The broader attack-search output is a separate setting.

TABLE S23. RELATIONSHIP TO EXISTING EVALUATION GUIDANCE. THESE ARE FIELD CORRESPONDENCES, NOT AN EMPIRICAL COMPARISON OF CHECKLISTS. ARP P-NUMBERS REFER TO SECTION 2 OF THE PUBLISHED PAPER [52].

Entries	Established obligation	Agent-specific specification used here
C1, C4	Realistic observer information and splits (Juarez et al.; Mathews et al.) [15], [16].	Separate transit, recipient and active access; identify supplied stage boundaries and association across model/tool links.
C2, C3	Sampling and label validity (Arp P1–P2).	Distinguish requested intent, logged action and checked outcome; state exclusions, null responses and recoverable errors.
C4, C5	Suitable comparisons (Arp P6), changed conditions and adaptive evaluation [15], [16], [19].	Name composition, topology, scaffold or model changes; distinguish fixed algorithm from refitting on protected traffic.
C6	Protection overhead and realistic evaluation [32], [16]; deployment scope (Arp P9).	Bind original-task utility to executed cover, and bytes/latency to the same protection condition and population.
C7	Application-relevant metrics and prevalence (Arp P7–P8).	Report event detection separately from supplied-unit classification; attach uncertainty and base rates to the named monitoring target.
C8	Evidence record supporting the correspondences above.	Give source version, result locator and accessible artifacts; distinguish source inspection from experimental reproduction.

reaching the planner; a hidden response is not a rollback. Compare protection, overhead and original-task completion on matched attempts, including rejected or interrupted attempts with their actual status. Any state or utility tolerance and any equivalence criterion must be specified before interpreting a small difference as preservation.

S8.4. Q4: discovery, association and inference

Record live concurrent execution and retain connection/stream association only when the chosen observer can obtain it. Keep complete-task and visible-tool intervals distinct. Use ground-truth clocks and event logs for evaluation, not to supply test-time task boundaries. Select detection thresholds on separate workloads or loads; report precision–recall and false alarms per time alongside correctly localized classification. Include a tool-only observer and an observer that can associate model and tool traffic. Section S10 tests live concurrent execution with varying session composition, empty sessions and repeated classes on one host. Its IoU sensitivity and constant controls test whether a detector resolves instances or merely covers coarse activity. The earlier fixed-composition study is in the artifact archive. Transfer across workload, load and production background traffic remains open; the offline compositions are a separate detector diagnostic.

S9. Deployment and Defense Source Anchors

The main deployment map separates the observer's rights from the tested source view. Table S26 supplies the protocol layer behind that map. TLS 1.3 record protection follows RFC 8446, Section 5 [25]. HTTP/2 concurrent streams and DATA framing follow RFC 9113, Sections 4–6; QUIC streams and migration follow RFC 9000, Sections 2 and 9; HTTP/3 maps requests to QUIC streams in RFC 9114, Sections 4 and 6. The MCP reference is the 2025-06-18 Streamable HTTP specification, including POST/GET, optional SSE and local stdio. These specifications do not prescribe a privacy padding or pacing policy.

TABLE S24. COMPLETED FORMS FOR ALL EIGHT REPORTING FIELDS. VERSIONS AND SOURCE LOCATIONS ARE GIVEN IN C8; ENTRIES PRESERVE THE SOURCE-SPECIFIC OBSERVER AND DENOMINATOR.

Field	WRA: decoy recovery	Communication graphs: live OHTTP	ANT: benchmark risk
C1	Reported: research-task domain mixture; passive observation and supplied task boundary.	Reported: live link capture with relay/trust assumptions; supplied workflow.	Reported: passive packet metadata with supplied episode boundaries.
C2	Reported: original query meaning is the target. Application authorization is not applicable.	Reported: one of six workflow labels. Application authorization is not applicable.	Partial: benign/misuse task labels are explicit; per-execution principal/approval ground truth is not located.
C3	Partial: real-plus-decoy research is specified; matched original-task outcome and execution-failure handling are not located.	Partial: the live population is distinct; matched task-goal outcome and failure selection are not located.	Partial: 20 positive and 189 negative test episodes; refusal-only Attack Code Generation is excluded, without general failure stratification.
C4	Partial: a supplied-task decoy comparison; the mapped record does not supply a complete split-unit description or a deployment-transfer test.	Reported: fold-wise classification within each live protected capture; this six-class population is distinct from the nine-class composition test.	Reported: within-task episode split and 209 test episodes; this is not a new-task or authorization-transfer test.
C5	Partial: test-side decoy mixing is checked; matched protected ICL examples/refitting remain unresolved.	Partial: fixed-family RF refitting is checked; the broader attack-search output is not bound to this live setting.	Not applicable: protected-condition adaptation, because this risk setting has no privacy intervention. Thirteen unprotected classifiers are reported.
C6	Partial: five extra research traces and recovery .770 to .565. Matched decoy bytes, delay and original-task utility are not located.	Partial: accuracy .30 and about 13-fold bytes/11-fold latency share the live condition; matched task utility is not located.	Not applicable: a protection/cost comparison. The setting evaluates benchmark-risk recognition.
C7	Partial: functional recovery is recorded; a matched uncertainty interval is not transcribed in this setting record. Deployment alarm prevalence is not this reconstruction metric.	Partial: six-class accuracy and cost ratios are recorded; a matched uncertainty interval is not transcribed in this setting record.	Reported: class totals and rounded accuracy, F1, FPR and recall. This SoK adds conditional binomial reference bands; raw clustering and deployment shift remain unknown.
C8	Reported: official USENIX 2026 PDF, Sec. 8.1, decoy Table 8; visibility Table 11 is separate. Selected implementation checks are in Section S4.	Reported: arXiv 2606.07150v3, Sec. 9.7 and the selected live evaluation path in Section S4; no independent rerun.	Reported: arXiv 2610.06514v1, Secs. 3.1, 4.1–4.2, App. A.1, Table 3. Public-table reconstruction is in the artifact's `impact-analysis/` directory.

The survey comparison uses Yao et al. v3, Sections 3 and 6.1.2; Shahriar et al. v1, Sections 1, 3–5; and Ferrag et al. v1, Sections I, V-A.1 and VI. They contribute scope comparisons, not new direct-Agent measurement rows or new publication-status evidence. Their original versions and locators are recorded in the review package.

Table S28 preserves each source-specific residual, cost and adaptation condition. The design-space table in the main paper instead states enforcement choices and costs to measure. An unmeasured cost in this table is not assigned a value by that design analysis.

TABLE S25. SOURCE RELATIONSHIPS USED IN THE SYNTHESIS. EXECUTION CHANGES QUALIFY BOTH FEEDBACK EXPOSURE AND COVER EVALUATION. SETTING IDS REFER TO THE SOURCE MATRIX AND ITS VERSION-SPECIFIC PASSAGE LOCATORS.

Finding	Observed basis	Conditional implication and open test
Execution organization	CG-composition and CB-topology report different transfer changes. TR-scaffold concerns terminal model attribution; ATR-main uses active repairs. TR-within and target/outcome records distinguish excluded sessions from observable step errors.	Recurring-task privacy and changed-deployment monitoring need distinct tests. Q1 asks which feature families survive specified changes; error normalization is an untested defense hypothesis.
Distributed exposure	WRA-prompt, CB-task and ANT-model-scope concern different interfaces. PT-metadata/PT-content compare a restricted view with added arguments. Recipient and active settings require their stated rights.	Enumerate the representations available to the observer. Q2 isolates feedback; Q4 tests discovery and association. The local complete-model scores do not supply feedback attribution.
Cover and execution cost	WRA-decoy and WRA-visibility have different utility conditions. CG-live pairs protected refitting with transport cost. The paired feedback study reports phase and complete-execution cells separately.	Constrain original-task utility when cover becomes work. Q3 tests transport termination versus isolated execution; harmful writes are not attributed to WRA.

TABLE S26. PROTOCOL AND DEPLOYMENT MAP: VISIBLE SIGNALS, INFERENCE CONSEQUENCES AND PROTECTION CHOICES. SPECIFICATIONS ESTABLISH THE PROTOCOL BEHAVIOR; THE LAST TWO COLUMNS DERIVE THE EVALUATION IMPLICATIONS.

Deployment feature	Signal and observation rights	Consequence for inference	Protection / measurement choice
Streaming model replies	SSE carries incremental events; encrypted sizes and gaps can reflect generation and buffering [20], [12].	An SSE event, TLS record and IP packet need not coincide. A front end can repacketize an upstream response.	Measure after the last visible repacketizer. Compare size padding with batching or pacing; price first-response and completion delay.
HTTP/2 connection reuse	Concurrent request streams share a TCP connection; stream IDs are inside encrypted HTTP/2 frames [24].	A visible five-tuple identifies a connection, not an Agent task. Shared scheduling and TCP loss affect several requests.	Record negotiated transport, concurrent streams and connection policy. Retain five-tuples in attack features without granting decrypted stream IDs.
HTTP/3 over QUIC	Streams share a QUIC connection; stream-level delivery reduces cross-stream blocking, and connection migration can change the five-tuple [39], [40].	HTTP/2 timing and TCP-flow segmentation results need a new transport test. Multiplexing mixes traffic without guaranteeing anonymity.	Evaluate at the observer's packet layer, including migration and loss; distinguish a connection from a fixed address pair.
MCP tool transport	Streamable HTTP uses POST/GET and optional SSE; local `stdio` has no network tool leg [23].	Remote tool requests, model requests and any downstream service calls expose different paths. MCP itself does not require HTTP/2.	State the MCP version and actual underlying transport. Examine both tool and model connections when results re-enter the prompt.
API gateway / relay	TLS termination, buffering, routing and shared upstream connections create new observation cuts [20], [3].	A transit attacker sees the gateway's output; a terminating gateway can inspect content. Upstream shaping may change downstream.	Put shaping at the last observed hop, or verify its survival. State which relay must be trusted and which party can link user and destination.
Shared serving state	Prefix/KV-cache reuse changes response timing under the service's sharing policy [11], [21].	An active querying tenant has a different attack interface from a passive packet observer. Padding bodies leaves cache timing separate.	Bind measurements to tenant isolation, cache scope and scheduling. Price reduced reuse in compute and latency.

TABLE S27. RELATIONSHIP TO BROADER SURVEYS, COMPARED BY THEIR STATED ORGANIZING QUESTIONS.

Survey	Organizing question	Focus added here
Yao et al. [53]	LLMs as security tools, attack tools and vulnerable systems; system side channels in Sec. 6.1.2	Bind a communication signal to observer access, execution target and measured population.
Shahriar et al. [54]	Agent applications, threats and defenses; architecture and evaluation trends	Trace task information across separately controlled services and compare transfer conditions.
Ferrag et al. [55]	Input, model, system/privacy and protocol attacks, including serving and MCP threats	Connect protocol representation, inference evidence and matched protection–utility costs.
WF evaluation SoK [16]	Attack/defense evaluation and its assumptions	Apply those obligations to feedback, tool effects and model/tool connection sharing.

TABLE S28. DEFENSE FIELD MAP. THE UPPER BLOCK PROTECTS INFORMATION ABOUT AN AGENT USER OR EXECUTION; THE LOWER BLOCK CHANGES IDENTITY ATTRIBUTION OR BLOCKS AN ADVERSARIAL AGENT. COSTS AND RESIDUALS REMAIN BOUND TO THEIR OWN CONDITION. NR MEANS NOT LOCATED AS A MATCHED MEASUREMENT IN THE INSPECTED PASSAGES, NOT ZERO COST.

Type / setting	Mechanism and measured residual	Cost / utility	Adaptive evaluation and remaining exposure
Execution cover: WRA [2, Sec. 8.1]	Five extra research traces: functional recovery .770 → .565	Additional research; matched decoy bytes, delay and original-task utility NR	Test-trace mixing located in code; matched protected ICL examples/refit unresolved. Domains still expose query information.
Association / visibility: WRA [2, Sec. 8.2]	100% → 10% domain visibility: recovery .770 → .610	Same sweep: report utility 8.55/8.54. Full live rerouting cost NR	Visibility sweep, not a matched adaptive attack search; remaining domains still leak. This utility does not price the decoy row.
Transport / cover: communication graphs [3, Secs. 9.3,9.8]	Metadata transformations; refitted and stronger capture-based attacks	Transform-based byte/delay costs; original-task completion NR	Adaptation evaluated on transformed captures. This is a different execution path from the live row below.
Association plus cover: live OHTTP [3, Sec. 9.7]	Six-class inference accuracy .30	About 13-fold bytes, 11-fold latency; matched task utility NR	Fixed-family RF refit on each protected capture confirmed in code; broader attack search not bound to this live result.
Identity obfuscation: AgentTrace [47, Sec. 4.4]	Identity prompts / temperature; attribution above 94% across prompt settings	Matched end-to-end utility/token/latency cost NR	Active designed probes tested; protected-condition retraining not located. Parser and repair behavior remains available.
Identity obfuscation: web-agent fingerprinting [7, Secs. 6–7]	Stealth/undetected agent configurations; site-side network and browser fingerprints	Benign-user utility not applicable; configuration cost NR	Contacted site observes both configurations; stealth modes often increase rather than reduce detectability.
Access blocking: web-agent study [7, Secs. 5–8]	Site anti-bot challenges; per-agent access outcomes	Form submission is the endpoint; benign-user privacy/utility is different	Stealth configurations evaluated. Blocking success neither conceals task information nor proves correct authorization.
Environment deception: CHeaT [10, Secs. 6.2,7]	Planted cloaks/traps; 0/33 defended CTF attempts reach initial discovery	10 planted data points per machine; benign-user matched cost NR	Separate defense-aware tests: informed prompt lowers DSR to 1.04% with 94.12% payload-detection FPR. Adaptation trades evasion against rejecting clean input.

S10. Randomized feedback and concurrent execution study

The paired design tests how the information forwarded after a tool response changes the model-link observation. It also measures the consequences of sharing connections and adding model-link shaping. The synthetic business actions execute in isolated fixtures; the tools are served by the MCP SDK and the workflow is executed by LangGraph. The earlier serial padding, common-prefix and fixed-composition studies are indexed in the artifact archive; their populations are not combined with this study.

S10.1. Task population and assignment

Six actions are available: refund, address update, escalation, replacement shipment, cancellation and receipt. Each task has one to three prerequisites sampled from account, policy, approval and status checks. A task assigned a temporary check error requires a single retry after receiving that terminal response. A correct business action changes the corresponding fixture field. The evaluator records the requested action, applied action, prerequisite completion and final state separately.

The local cohort contains 312 task instances in 64 sessions per model and condition. The training, validation and test partitions contain 32, 8 and 24 sessions, respectively, with 156, 36 and 120 tasks. Each class has 26 training, six validation and 20 test instances. Class counts vary within sessions: the complete cohort contains ten empty sessions, 41 sessions with repeated classes and 59 missing at least one class. Four test sessions are empty, leaving 20 nonempty clusters for supplied-window test accuracy. The test partition uses four held-out request phrasings per action. Sessions vary in arrivals and contents but do not constitute independent application domains.

Arrivals follow the frozen session schedule generated with exponential interarrival times at rate .55/s. Idle periods of 3–7 seconds precede and follow the work; an empty session lasts 12 seconds. Independent MCP `tools/list` background requests follow a .35/s exponential process. These are live SDK requests on the captured service, not a sample of enterprise background traffic. A session has 0–10 task arrivals; the concurrent conditions use up to four workers. Condition order rotates by session; the local models run as separate cohorts. Empty sessions remain in duration and false-alarm denominators.

S10.2. Feedback isolation and intervention

The initial instruction, system message and admissible initial action are identical across tasks. The fixture becomes available only after the initial model response terminates. The first tool response then contains the customer request, required checks, order context and three to seven supplementary records. The records use service-specific schemas with randomized content; they deliberately provide a recoverable dependency to test, rather than estimate the frequency of such schemas in deployed services.

Full forwards that response to the next model request. *Projected* applies a fixed allowlist after receiving the same complete tool response. It retains `customer_request`, `required_checks`, order reference, amount and address, plus check/result fields, while removing `supporting_records`. Task seeds and tool fixtures are paired across conditions. The audit checks equality of fixture hashes, equality of unprojected fields, and the exact allowlist operation. Projection preserves the business information needed by the designed task; measured completion determines whether the model still uses it successfully.

Both serial conditions pad tool requests to a 4,096-byte quantum and responses to a 16,384-byte quantum. JSON whitespace and SSE-comment padding are removed by parsing before the structured tool result is forwarded. *Dedicated* uses separate connections for each worker and service; *Shared* uses a single connection per service. *Joint* adds model-request padding to a 32,768-byte quantum and emits model SSE frames of 512 bytes every 25 ms for at least 40 frames. Responses extending beyond the minimum schedule remain extended and are counted as overruns. Padding quanta can be exceeded. Thus the intervention is an implemented shaping policy with measured residual signals, not a constant-time channel.

S10.3. Execution and transport

LangGraph 1.0.3 [56] controls alternating model/tool nodes. MCP SDK 1.21.1 serves Streamable HTTP through Hypercorn 0.17.3. Local model requests go through a TLS 1.3/HTTP/2 gateway that forwards incremental Ollama output as SSE. Local model identities and weights are fixed for Qwen3:8B and Llama3.1:8B. The model chooses among unfinished prerequisites and then all six business actions; the host guard enforces protocol prerequisites without supplying the correct action label. The eight-step cap and functional errors remain part of the outcome population.

Readiness tasks are excluded from the formal partitions. The initial readiness run exposed premature commitments and repeated ticket retrieval; the documented amendment added state-dependent admissible-action constraints before the formal cohort was frozen. A second readiness run checks that the workflow and both models can complete the tasks. Readiness outcomes are preserved independently of formal attack scores.

Capture retains encrypted TCP payload size, relative timestamp, direction and physical connection grouping. Packet features never use a decrypted HTTP/2 stream identifier, task-to-stream mapping, tool plaintext or requested class. A separate private wire check uses TLS keys to verify TLS 1.3, HTTP/2, overlapping streams and incremental delivery. Those verification inputs are excluded from the attack representation.

The formal local capture contains 811,401 payload-bearing TCP packets across 640 sessions. Request and state

checks account for all 3,120 executions and 13,680 terminal model requests. Wire verification finds 2,048 TLS 1.3 negotiations selecting HTTP/2 and 39,854 closed HTTP/2 request streams. All model responses contain at least two nonempty DATA frames with a positive first-to-last span. In each shared condition, 107 of 128 sessions contain overlapping model requests on one connection, reaching four active streams. Serial and dedicated conditions have at most one active model request per connection. These checks establish the implemented protocol and overlap; the capture does not provide a packet-drop counter.

S10.4. Recognition models and selection

Supplied-window attribution has three units: the initial model call, the first model call after the ticket, and the complete execution. Prefix and post-ticket attribution use model-link metadata; complete execution considers the tool link, model link and joint view. A window includes all observed connections in its interval, so concurrent observations include other active tasks. Phase attribution is evaluated for serial Full/Projected, as specified by the mechanism contrasts. Shared/Joint are evaluated through complete windows and continuous localization. Their post-ticket timestamps exist for ground truth, but they do not identify a transit-visible per-task stream: a phase score would mix concurrent tasks. No Joint post-ticket outcome is claimed. Missing phases in declared comparisons remain represented instead of being excluded.

The primary models are standardized logistic regression (LR) and the released 175-feature k-FP extractor [46] with a 1,000-tree random forest. LR selects $C \in \{.1, 1, 10\}$ by validation accuracy, breaking ties toward smaller C. The k-FP implementation is version-pinned; Python iterator compatibility changes leave its features unchanged. Each encrypted TCP-payload packet supplies a signed event and relative time to that extractor. Traces with fewer than three packets, only one direction or zero duration receive a flagged zero feature vector. Joint k-FP features concatenate per-link vectors. The closed-world forest evaluation is distinct from k-FP's open-world decision procedure.

Additional attacks are a 500-tree forest on 33 aggregate features and a temporal CNN. The CNN receives at most 512 ordered packets (first 384 and final 128 when truncated), represented by signed log size, log interarrival time, log elapsed time, service role and validity mask. Three convolutional layers with kernel width five precede pooled and fully connected layers. Channel width is selected from 32 and 64; Adam uses learning rate .001, weight decay .001 and batch size 32 for up to 40 epochs. Validation accuracy, then loss and earlier epoch, determines the checkpoint. This is a temporal baseline, not a reproduction of DF [37] or Var-CNN [38]. Its low held-out scores do not establish training failure; the saved selections and outcomes remain reported with the other attacks.

All validation selections are saved before any test score is computed. Each condition is fitted separately; a second evaluation transfers the fitted Full predictor unchanged. Both results are retained. The majority baseline is determined by training labels. Test accuracy, macro-F1, complete confusion matrices and paired correctness are released for every declared learner and view.

S10.5. Continuous localization and uncertainty

Continuous detectors receive session boundaries and connection identities but no task boundaries or private stream assignment. A ground-truth execution interval starts at worker admission and ends at its terminal outcome; a scheduled task waiting for a worker has not yet entered that interval. Six binary LR models and a multilabel forest use .25-second grids with 1-, 2- or 4-second lookback. Validation event F1 at intersection-over-union (IoU) .5 selects the lookback and threshold, with ties favoring a higher threshold and shorter lookback. Threshold candidates span .1–.9. Gap detection with a supplied-window classifier and constant class-presence outputs provide additional controls.

Predicted events are matched one-to-one to true task instances using maximum-cardinality matching subject to the IoU threshold, with overlap as a tie-break. Duplicate same-class instances remain distinct truths. Binary class-activity models can merge overlapping instances; the missed-instance penalties are retained. We report IoU .3/.5/.7, precision, recall, F1, mean false events per session, duration-normalized false events per hour, matched endpoint errors and empty-session alarms. A strong supplied-window score is not substituted for this event-level evaluation.

Paired uncertainty resamples complete test sessions 2,000 times. LR additionally uses 300 paired training/test session resamples and refits the predictor, keeping its validation-selected hyperparameter fixed. These intervals cover the stated resampling populations, conditional on the designed workload and selected model family. The two feedback contrasts, two primary attacks and two local execution models form one eight-comparison Holm family. Session-level paired sign permutations use 10,000 draws on saved predictions, so their p-values condition on the fitted attacks; within-session label permutations use 2,000 draws. A ten-percentage-point contrast was declared practically relevant before scoring. No prospective power calculation established that the twenty nonempty test clusters could reliably detect that contrast; the earlier common-prefix study has a different population and cannot supply that guarantee. RF/CNN and deployment comparisons remain exploratory; an interval crossing zero does not establish equivalence.

A post-scoring numerical audit enumerates the same session-sign distribution exactly for every declared primary contrast. It retains the original Monte Carlo values, predictions, clusters and multiplicity families. For the smallest local adjusted value, the approximation is .049595 and the exact value is .050781. This threshold sensitivity precludes treating that comparison as a robust rejection. The artifact provides integer numerators, denominators and an independently checked enumeration routine.

Resource accounting separates active execution from session occupancy. Task time runs from worker admission

to completion and excludes waiting for an available worker. Session time spans connection setup, arrivals, execution, background traffic and idle margins; its sum divided by the task count gives an amortized session cost, not a per-task response latency. Each captured packet is charged once to its session, including empty sessions. Paired session resampling preserves concurrent workloads without summing overlapping task windows.

S10.6. Hosted retest and review boundary

The bounded remote plan uses the same workflow and paired Full/Projected intervention with an authorized account and synthetic fixtures. Its separate 96-task population has 36 training, 12 validation and 48 test tasks per condition across 30 sessions. Eighteen test sessions include two empty sessions, leaving 16 nonempty test clusters. Training and validation sessions contain the remaining 36 and 12 tasks. The hosted model identifier is a mutable alias. Fixture seeds remain paired, but provider requests do not enforce a generation seed. Capture observes the local tunnel path plus the local tool service. This retest changes the backend and network together. Provider research permission and terms applicability remain unverified. A missing response terminal is retained as unknown and stops collection without automatic replay.

All 192 formal remote attempts have terminal outcomes, accounting for 800 model requests. Five invalid model outputs terminate their tasks and remain in the intended-label denominator. Full completes 92 of 96 goals and Projected 95 of 96; the held-out counts are 46/48 and 47/48. Wire checks verify TLS 1.3/HTTP/2 on the captured services, 112 negotiations selecting HTTP/2 and 3,196 closed request streams. Every model response has multiple nonempty DATA frames with distinct first and last timestamps. These checks concern the observed protocol, not a fixed provider model or physical-NIC timing guarantee.

The remote test schedule was amended before remote requests or formal attack scoring. The same 48 test instances were redistributed from seven to sixteen nonempty sessions, retaining their labels, fixtures, seeds and partition. With seven clusters, the smallest exact two-sided sign-test value is $2/2^7$, whose first four-comparison Holm step exceeds .05. Sixteen clusters remove that resolution constraint; they do not guarantee power for a ten-point effect. The artifact preserves both plans and the amendment. Local collection, attack selection, multiplicity and the remote request budget remain fixed.

The offline review package includes payload-free relative packet metadata, designed cohort specifications, features, predictions and outcome records. Credentials, TLS keys, raw captures and transcripts remain private. Private wire receipts document checks on excluded inputs; public metadata checks provide the reproducible analysis path. Collection and analysis snapshots, hashes and implementation amendments are stored with the artifact rather than embedded in the scientific narrative.

S10.7. Verified outcomes and all primary comparisons

The formal cohorts contain 3,120 local and 192 remote attempts, all with terminal records. Local business-goal completion is 1,555/1,560 for Qwen3:8B and 1,542/1,560 for Llama3.1:8B. Remote completion is 187/192. Wrong actions and invalid model outputs remain in the inference population. Table S35 separates all-run completion from held-out completion and reports service time, session time and bytes.

Table S32 contains every primary phase and forwarding comparison. The local LR Full–Projected point differences are 16.7 and 15.0 percentage points, with unadjusted training/test-refit intervals [4.0,29.8] and [3.1,32.8] points. All exact Holm-adjusted primary values exceed .05. The remote LR difference is 10.4 points with interval [-3.2,41.5]; its k-FP difference reverses direction. The intervals and permutation tests condition on different fitted/resampled quantities; the unadjusted intervals do not override the complete multiplicity-adjusted tests.

Table S34 retains the original approximation beside exact enumeration. Table S36 gives every matched and frozen recognition outcome, including the exploratory random forest and CNN. Table S31 places complete-execution outcomes beside prefix and first-feedback outcomes. Model-only LR changes from 46/120 to 44/120 for Qwen, 62/120 to 33/120 for Llama, and 9/48 to 9/48 for the hosted model. Complete joint-view counts are 54 to 39, 53 to 29 and 12 to 6, respectively. These views include later requests, responses and decisions. Their heterogeneity rules out characterizing the whole intervention using only its first-feedback point difference. The model's responses and retained task fields are possible sources of residual recognition, but the current bidirectional representation does not attribute that recognition to either one.

In the local Shared joint view, LR and forest event detectors obtain 35.0–43.3% recall with 4.3–18.6% precision at IoU .5. The complete event table includes each condition, permitted view, baseline and empty-session count. Mean false events per session divides the aggregate false count by all 24 local test sessions, including four empty sessions (18 and two remotely). Hourly rates remain normalized by the corresponding recorded duration. Session lengths and task loads define these denominators; neither unit supplies enterprise alarm performance. A deployment-prior adjustment would require an event-arrival model and false-alarm process, rather than substituting a classification prior into these event metrics. Table S33 places IoU .3/.5/.7 side by side for shared and jointly shaped connections. At .3, the all-active Shared F1 (.295/.508 for Qwen/Llama) exceeds both learned detectors; at .5, the ordering reverses. The same predictions are scored at every threshold. The artifact contains all views, per-session predictions and intervals. A loose threshold can reward coarse occupancy even with random task composition; the .5 primary endpoint remains unchanged.

Projection's local model-byte ratio is approximately .78. Joint shaping increases model-link bytes by 19.5 and 22.5 times relative to Shared; 1,818 of 2,736 shaped model responses exceed the forty-frame minimum. The remote Projected/Full byte ratio is .932, with paired interval [.868,1.003]. These costs and the unequal recognition changes distinguish a measured shaping policy from a confidentiality guarantee.

Independent offline checks reconstruct all packet-derived aggregate and sequence features, fixture outcomes, LR selections and recognition arithmetic, event matching and cost accounting. Saved LR/forest inference is replayed with separate array arithmetic; CNN replay uses its exported tensors and the same numerical library. The package records the checking scope, implementation corrections and private raw-capture verification separately.

S10.8. Saved prefix and execution-path diagnostics

Table S29 exposes the prefix permutation results already computed in the original analysis. Qwen Full LR's 25/120 has unadjusted $p = .0595$; Llama's 19/120 has $p = .6092$. The within-session permutation preserves each session's label composition and fixes the predictor. These diagnostics neither demonstrate an information-free prefix nor change the primary post–prefix family.

Table S30 audits existing execution records, pairing held-out tasks by executor, session and unit. Each Full/Projected local arm has 518 executed tool actions, and all 120 pairs have equal call counts. Each arm also has 26 extra prerequisite-check calls. Exact action sequences agree in 85 Qwen and 94 Llama pairs; unordered action multisets agree in 118 and 119 pairs. Applied business actions agree in the same respective counts. Hence order changes and a few changed decisions remain even though call depth and repeated-check totals are matched. These summaries cannot identify the contribution of message contents, encoded size, action order or response timing to the recognition contrast.

What a separating design must preserve. A future factorial contrast would need task-valid supplementary-content replacement and neutral padding matched at the observed encoding layer. Changing a record's category can change its semantics; matching JSON bytes need not match token counts, responses or timing. Record-count sweeps must account for their simultaneous size and workload changes. Public tool schemas could identify candidate fields, while prevalence of realized leakage would require sampled executions and an explicit denominator.

A new confirmation cohort would fix its estimand, validation-based attacker selection and multiplicity family before collection. Planning should vary plausible effects and session dependence; the pilot's fitted predictions alone do not establish the required sample size. Pooling models would define a weighted, model-stratified effect and retain each stratum's result. The existing LR/k-FP tests and exploratory RF/CNN results keep their original roles.

TABLE S29. SAVED PREFIX DIAGNOSTICS. THESE ARE THE ORIGINAL 2,000 WITHIN-SESSION LABEL PERMUTATIONS ON FIXED PREDICTIONS, WITHOUT MULTIPLICITY ADJUSTMENT. THEY TEST ASSOCIATION CONDITIONAL ON THE OBSERVED SESSIONS; A LARGE VALUE DOES NOT ESTABLISH AN INFORMATION-FREE PREFIX.

Executor	Policy	n	LR correct	LR p	k-FP correct	k-FP p
Qwen3:8B	Full	120	25	0.0595	18	0.8071
Qwen3:8B	Projected	120	23	0.3998	23	0.4473
Llama3.1:8B	Full	120	19	0.6092	14	0.8646
Llama3.1:8B	Projected	120	16	0.5347	20	0.3988
Hosted alias	Full	48	5	0.5412	3	0.9960
Hosted alias	Projected	48	5	0.8141	6	0.9150

TABLE S30. DESCRIPTIVE AUDIT OF THE 120 HELD-OUT FULL/PROJECTED TASK PAIRS PER LOCAL EXECUTOR. CALLS ARE EXECUTED TOOL ACTIONS. EXTRA CHECKS COUNT REPEATED ACCOUNT/POLICY/APPROVAL/STATUS CALLS BEYOND THE FIRST. THE LAST FOUR COLUMNS COUNT AGREEING PAIRS; A MULTISET IGNORES CALL ORDER. THESE SUMMARIES DO NOT ASSIGN THE RECOGNITION CHANGE TO A MEDIATOR.

Executor	Total calls F/P	Extra checks F/P	Same count	Same sequence	Same multiset	Same applied action
Qwen3:8B	518/518	26/26	120	85	118	118
Llama3.1:8B	518/518	26/26	120	94	119	119

TABLE S31. CONDITION-MATCHED RECOGNITION: CORRECT PREDICTIONS OUT OF n (SIX-CLASS REFERENCE $n/6$). PREFIX IS THE INITIAL FULL MODEL CALL; POST IS ITS FIRST CALL AFTER THE TICKET. COMPLETE SPANS ALL REQUESTS AND RESPONSES. RF AND CNN ARE EXPLORATORY (E); NEITHER REPLACES LR/K-FP IN THE PRIMARY FAMILY. p_{proj} IS THE EXACT HOLM-ADJUSTED FULL–PROJECTED POST TEST. COMPLETE-VIEW COMPARISONS ARE DESCRIPTIVE. SUPPLEMENT S1 REPORTS ALL UNCERTAINTY AND FROZEN-TRANSFER RESULTS.

Executor	Attack	n	Prefix	Post M		Complete M		Complete T+M		p_{proj}
				Full	Proj.	Full	Proj.	Full	Proj.	
Qwen3:8B	LR	120	25	43	23	46	44	54	39	0.0778
Qwen3:8B	kFP	120	18	32	20	43	39	39	30	0.1989
Qwen3:8B	RF (E)	120	25	50	30	61	31	55	30	—
Qwen3:8B	CNN (E)	120	20	20	20	22	22	20	22	—
Llama3.1:8B	LR	120	19	40	22	62	33	53	29	0.1172
Llama3.1:8B	kFP	120	14	31	20	47	34	45	31	0.1989
Llama3.1:8B	RF (E)	120	24	59	19	73	34	55	30	—
Llama3.1:8B	CNN (E)	120	20	26	20	22	15	19	18	—
Hosted alias	LR	48	5	15	10	9	9	12	6	0.9214
Hosted alias	kFP	48	3	6	7	9	5	13	11	1.0000
Hosted alias	RF (E)	48	8	15	9	7	7	13	8	—
Hosted alias	CNN (E)	48	8	11	5	10	8	9	11	—

TABLE S32. PAIRED FEEDBACK-PATH RESULTS. ALL SCORES ARE CORRECT INTENDED-ACTION PREDICTIONS IN THE MODEL-LINK VIEW. MODELS ARE FITTED WITHIN EACH CONDITION. FULL MINUS PROJECTED IS A PERCENTAGE-POINT DIFFERENCE; LR INTERVALS RESAMPLE TRAINING AND TEST SESSIONS WITH REFITTING, WHILE K-FP INTERVALS CONDITION ON FITTED PREDICTORS. THE ADJUSTED p COLUMNS ENUMERATE THE SESSION-SIGN TESTS ON SAVED PAIRED PREDICTIONS EXACTLY; EIGHT LOCAL TESTS AND FOUR REMOTE TESTS FORM SEPARATE HOLM FAMILIES. THE DECLARED MONTE CARLO APPROXIMATION IS RETAINED BESIDE THE EXACT VALUES IN THE SUPPLEMENT. HOSTED RESULTS USE A SEPARATE COHORT AND MUTABLE ALIAS.

Executor	Attack	Prefix	Full post	Projected post	Full minus Projected [95% CI]	p_{post}	p_{proj}
Qwen3:8B	LR	25/120	43/120	23/120	$+16.7$ $[+4.0, +29.8]$	0.1357	0.0778
Qwen3:8B	kFP	18/120	32/120	20/120	$+10.0$ $[-1.6, +20.5]$	0.1172	0.1989
Llama3.1:8B	LR	19/120	40/120	22/120	$+15.0$ $[+3.1, +32.8]$	0.0508	0.1172
Llama3.1:8B	kFP	14/120	31/120	20/120	$+9.2$ $[+0.0, +17.4]$	0.1357	0.1989
Hosted alias	LR	5/48	15/48	10/48	$+10.4$ $[-3.2, +41.5]$	0.1562	0.9214
Hosted alias	kFP	3/48	6/48	7/48	-2.1 $[-17.4, +15.4]$	1.0000	1.0000

TABLE S33. EVENT OVERLAP SENSITIVITY ON THE SAME PREDICTIONS. EACH CONDITION CONTAINS 120 TASKS IN 24 SESSIONS, INCLUDING FOUR EMPTY SESSIONS. LEARNED DETECTORS USE T+M; ALL-ACTIVE EMITS ONE SESSION INTERVAL PER CLASS. IOU .5 IS PRIMARY; .3/.7 ARE SENSITIVITY CHECKS. FP/SESSION DIVIDES ALL FALSE EVENTS BY 24, RETAINING EMPTY SESSIONS. SUPPLEMENT S1 RETAINS HOURLY NORMALIZATION, COUNTS AND INTERVALS FOR THESE BOUNDED RECORDINGS.

Executor	Policy	Detector	F1 (.3)	F1 (.5)	F1 (.7)	P (.5)	R (.5)	FP/session (.5)
Qwen3:8B	Shared	LR	0.216	0.119	0.048	0.072	0.350	22.54
Qwen3:8B	Shared	RF	0.106	0.077	0.035	0.043	0.400	44.79
Qwen3:8B	Shared	all-active	0.295	0.038	0.000	0.035	0.042	5.79
Qwen3:8B	Joint	LR	0.196	0.117	0.050	0.070	0.375	25.04
Qwen3:8B	Joint	RF	0.329	0.188	0.090	0.123	0.400	14.29
Qwen3:8B	Joint	all-active	0.265	0.038	0.000	0.035	0.042	5.79
Llama3.1:8B	Shared	LR	0.333	0.216	0.104	0.144	0.433	12.88
Llama3.1:8B	Shared	RF	0.464	0.249	0.105	0.186	0.375	8.21
Llama3.1:8B	Shared	all-active	0.508	0.098	0.000	0.090	0.108	5.46
Llama3.1:8B	Joint	LR	0.441	0.251	0.104	0.186	0.383	8.38
Llama3.1:8B	Joint	RF	0.429	0.274	0.129	0.199	0.442	8.92
Llama3.1:8B	Joint	all-active	0.530	0.121	0.000	0.111	0.133	5.33

TABLE S34. NUMERICAL PRECISION SENSITIVITY FOR THE COMPLETE DECLARED PRIMARY FAMILIES. MC RETAINS THE PRESPECIFIED 10,000-DRAW SESSION-SIGN APPROXIMATION; EXACT ENUMERATES THE SAME SIGN-FLIP DISTRIBUTION USING INTEGER COUNTS. THIS POST-SCORING ARITHMETIC AUDIT CHANGES NO PREDICTIONS, CLUSTERS, HYPOTHESES OR MULTIPLICITY FAMILIES. A PASS AUDIT MEANS ARITHMETIC CONSISTENCY, NOT HYPOTHESIS REJECTION.

Executor	Attack	Contrast	MC p	Exact p	MC Holm	Exact Holm
Llama3.1:8B	LR	Post–prefix	0.006199	0.006348	0.049595	0.050781
Llama3.1:8B	kFP	Post–prefix	0.031997	0.033936	0.127987	0.135742
Llama3.1:8B	LR	Full–Projected	0.021198	0.021362	0.109189	0.117188
Llama3.1:8B	kFP	Full–Projected	0.096890	0.099457	0.193781	0.198914
Qwen3:8B	LR	Post–prefix	0.036796	0.035339	0.127987	0.135742
Qwen3:8B	kFP	Post–prefix	0.018198	0.019531	0.109189	0.117188
Qwen3:8B	LR	Full–Projected	0.010499	0.011108	0.073493	0.077759
Qwen3:8B	kFP	Full–Projected	0.141986	0.144592	0.193781	0.198914
Hosted alias	LR	Post–prefix	0.039096	0.039062	0.156384	0.156250
Hosted alias	kFP	Post–prefix	0.558044	0.562500	1.000000	1.000000
Hosted alias	LR	Full–Projected	0.308369	0.307129	0.925107	0.921387
Hosted alias	kFP	Full–Projected	1.000000	1.000000	1.000000	1.000000

TABLE S35. BUSINESS COMPLETION AND RESOURCE ACCOUNTING. ALL-RUN AND HELD-OUT SUCCESS USE DISTINCT DENOMINATORS. REMAINING COLUMNS DESCRIBE THE HELD-OUT POPULATION. MEDIAN AND P95 TASK TIMES START AT WORKER ADMISSION AND EXCLUDE WORKER QUEUEING. SESSION SECONDS/TASK AMORTIZE COMPLETE SESSION TIME, INCLUDING SETUP, BACKGROUND AND IDLE MARGINS; THEY ARE NOT PER-TASK RESPONSE LATENCIES. BYTES COUNT EACH PACKET ONCE PER SESSION. LOCAL AND HOSTED ABSOLUTE COSTS BELONG TO DIFFERENT SYSTEMS.

Executor	Condition	All success	Test success	Median s	P95 s	Session s/task	T KiB/task	M KiB/task
Qwen3:8B	Full	311/312	119/120	3.20	8.06	7.14	152.4	13.1
Qwen3:8B	Projected	311/312	119/120	3.04	7.89	6.91	151.4	10.2
Qwen3:8B	Dedicated	311/312	119/120	11.01	23.40	6.76	173.8	14.2
Qwen3:8B	Shared	311/312	119/120	9.61	31.78	6.91	144.6	12.7
Qwen3:8B	Joint	311/312	119/120	9.22	26.73	6.82	149.0	247.6
Llama3.1:8B	Full	309/312	117/120	6.84	9.62	9.82	165.4	13.1
Llama3.1:8B	Projected	306/312	118/120	6.82	9.74	9.94	167.4	10.3
Llama3.1:8B	Dedicated	309/312	117/120	21.38	29.28	9.22	179.7	14.2
Llama3.1:8B	Shared	309/312	117/120	20.94	28.99	9.04	151.3	12.7
Llama3.1:8B	Joint	309/312	117/120	23.07	32.50	9.42	161.0	286.2
Hosted alias	Full	92/96	46/48	14.49	33.24	20.76	256.1	108.5
Hosted alias	Projected	95/96	47/48	13.83	28.93	20.42	253.6	101.1

TABLE S36: Complete recognition outcomes. Parentheses give the test denominator. Each attack cell gives correct predictions as matched fit / frozen Full fit; a dash means the same Full model is already reported. Prefix and post-ticket use M; complete-execution views are separately fitted. Every planned terminal outcome remains in the intended-label denominator.

Executor (n)	Condition	Phase	View	LR	k-FP	RF	CNN
Qwen3:8B (120)	Full	Prefix	M	25 / –	18 / –	25 / –	20 / –
Qwen3:8B (120)	Full	Post-ticket	M	43 / –	32 / –	50 / –	20 / –
Qwen3:8B (120)	Full	Complete	T	19 / –	21 / –	23 / –	20 / –
Qwen3:8B (120)	Full	Complete	M	46 / –	43 / –	61 / –	22 / –
Qwen3:8B (120)	Full	Complete	T+M	54 / –	39 / –	55 / –	20 / –
Qwen3:8B (120)	Projected	Prefix	M	23 / 24	23 / 19	21 / 16	20 / 20
Qwen3:8B (120)	Projected	Post-ticket	M	23 / 20	20 / 20	30 / 20	20 / 20
Qwen3:8B (120)	Projected	Complete	T	18 / 20	20 / 24	20 / 24	20 / 22
Qwen3:8B (120)	Projected	Complete	M	44 / 20	39 / 30	31 / 20	22 / 16
Qwen3:8B (120)	Projected	Complete	T+M	39 / 19	30 / 28	30 / 16	22 / 22
Qwen3:8B (120)	Dedicated	Complete	T	23 / 21	25 / 20	24 / 20	20 / 20
Qwen3:8B (120)	Dedicated	Complete	M	25 / 24	22 / 19	24 / 23	25 / 19
Qwen3:8B (120)	Dedicated	Complete	T+M	30 / 28	23 / 21	24 / 20	16 / 21
Qwen3:8B (120)	Shared	Complete	T	20 / 18	19 / 21	17 / 19	20 / 20
Qwen3:8B (120)	Shared	Complete	M	25 / 26	25 / 19	30 / 24	17 / 20
Qwen3:8B (120)	Shared	Complete	T+M	23 / 21	22 / 20	22 / 24	19 / 19
Qwen3:8B (120)	Joint	Complete	T	19 / 24	17 / 16	19 / 19	17 / 19
Qwen3:8B (120)	Joint	Complete	M	18 / 20	24 / 20	22 / 20	25 / 20
Qwen3:8B (120)	Joint	Complete	T+M	20 / 20	22 / 21	19 / 19	25 / 21
Llama3.1:8B (120)	Full	Prefix	M	19 / –	14 / –	24 / –	20 / –
Llama3.1:8B (120)	Full	Post-ticket	M	40 / –	31 / –	59 / –	26 / –
Llama3.1:8B (120)	Full	Complete	T	17 / –	18 / –	19 / –	17 / –
Llama3.1:8B (120)	Full	Complete	M	62 / –	47 / –	73 / –	22 / –
Llama3.1:8B (120)	Full	Complete	T+M	53 / –	45 / –	55 / –	19 / –
Llama3.1:8B (120)	Projected	Prefix	M	16 / 20	20 / 12	14 / 21	20 / 20
Llama3.1:8B (120)	Projected	Post-ticket	M	22 / 20	20 / 27	19 / 20	20 / 21
Llama3.1:8B (120)	Projected	Complete	T	18 / 22	19 / 20	16 / 23	19 / 17
Llama3.1:8B (120)	Projected	Complete	M	33 / 20	34 / 23	34 / 16	15 / 17
Llama3.1:8B (120)	Projected	Complete	T+M	29 / 20	31 / 27	30 / 19	18 / 20
Llama3.1:8B (120)	Dedicated	Complete	T	16 / 21	17 / 19	15 / 20	21 / 21
Llama3.1:8B (120)	Dedicated	Complete	M	17 / 24	13 / 16	13 / 20	19 / 18
Llama3.1:8B (120)	Dedicated	Complete	T+M	20 / 23	16 / 21	18 / 20	19 / 18
Llama3.1:8B (120)	Shared	Complete	T	26 / 19	13 / 15	22 / 19	16 / 20
Llama3.1:8B (120)	Shared	Complete	M	19 / 21	19 / 17	21 / 21	19 / 18
Llama3.1:8B (120)	Shared	Complete	T+M	19 / 22	17 / 17	17 / 21	20 / 20
Llama3.1:8B (120)	Joint	Complete	T	17 / 18	23 / 17	15 / 22	24 / 19
Llama3.1:8B (120)	Joint	Complete	M	25 / 20	22 / 20	16 / 20	21 / 20
Llama3.1:8B (120)	Joint	Complete	T+M	22 / 20	17 / 20	17 / 20	18 / 18
Hosted alias (48)	Full	Prefix	M	5 / –	3 / –	8 / –	8 / –
Hosted alias (48)	Full	Post-ticket	M	15 / –	6 / –	15 / –	11 / –
Hosted alias (48)	Full	Complete	T	8 / –	8 / –	10 / –	11 / –
Hosted alias (48)	Full	Complete	M	9 / –	9 / –	7 / –	10 / –
Hosted alias (48)	Full	Complete	T+M	12 / –	13 / –	13 / –	9 / –
Hosted alias (48)	Projected	Prefix	M	5 / 5	6 / 5	6 / 8	8 / 8
Hosted alias (48)	Projected	Post-ticket	M	10 / 7	7 / 6	9 / 7	5 / 11
Hosted alias (48)	Projected	Complete	T	5 / 8	10 / 7	8 / 8	7 / 13
Hosted alias (48)	Projected	Complete	M	9 / 7	5 / 10	7 / 6	8 / 11
Hosted alias (48)	Projected	Complete	T+M	6 / 6	11 / 8	8 / 8	11 / 13

TABLE S37: Complete event results at the primary IoU .5. P, R and F1 use one-to-one task-instance matching. Empty FP counts false events in empty test sessions. The artifact additionally provides .3/.7 overlap results, validation choices, per-session counts and uncertainty.

Executor	Condition	View	Detector	P	R	F1	FP/hour	Empty FP
Qwen3:8B	Dedicated	T	LR	0.080	0.292	0.125	1788.5	15
Qwen3:8B	Dedicated	T	RF	0.051	0.317	0.088	3119.8	70
Qwen3:8B	Dedicated	T	gap	0.060	0.058	0.059	483.7	16
Qwen3:8B	Dedicated	M	LR	0.028	0.250	0.051	4606.5	3
Qwen3:8B	Dedicated	M	RF	0.079	0.317	0.127	1961.5	6
Qwen3:8B	Dedicated	M	gap	0.041	0.075	0.053	923.1	4
Qwen3:8B	Dedicated	T+M	LR	0.033	0.300	0.059	4726.3	3
Qwen3:8B	Dedicated	T+M	RF	0.139	0.392	0.205	1291.4	0
Qwen3:8B	Dedicated	T+M	gap	0.056	0.042	0.048	372.8	16
Qwen3:8B	Full	T	LR	0.011	0.033	0.016	1563.4	2
Qwen3:8B	Full	T	RF	0.009	0.142	0.017	7673.9	60
Qwen3:8B	Full	T	gap	0.006	0.025	0.010	2118.1	30
Qwen3:8B	Full	M	LR	0.054	0.358	0.094	3168.7	0
Qwen3:8B	Full	M	RF	0.022	0.317	0.041	7136.0	0
Qwen3:8B	Full	M	gap	0.005	0.008	0.006	882.5	4
Qwen3:8B	Full	T+M	LR	0.049	0.383	0.087	3748.7	0
Qwen3:8B	Full	T+M	RF	0.036	0.408	0.067	5471.7	0
Qwen3:8B	Full	T+M	gap	0.011	0.025	0.015	1159.9	23
Qwen3:8B	Projected	T	LR	0.007	0.067	0.013	4986.1	56
Qwen3:8B	Projected	T	RF	0.010	0.117	0.018	6019.9	40
Qwen3:8B	Projected	T	gap	0.000	0.000	0.000	1120.6	23
Qwen3:8B	Projected	M	LR	0.047	0.208	0.077	2189.0	0
Qwen3:8B	Projected	M	RF	0.003	0.033	0.006	5355.3	0
Qwen3:8B	Projected	M	gap	0.015	0.025	0.019	833.9	4
Qwen3:8B	Projected	T+M	LR	0.045	0.300	0.079	3283.6	0
Qwen3:8B	Projected	T+M	RF	0.020	0.192	0.036	4881.9	2
Qwen3:8B	Projected	T+M	gap	0.008	0.017	0.010	1133.6	23
Qwen3:8B	Shared	T	LR	0.037	0.217	0.064	2901.0	86
Qwen3:8B	Shared	T	RF	0.049	0.367	0.087	3704.4	77
Qwen3:8B	Shared	T	gap	0.073	0.025	0.037	165.0	9
Qwen3:8B	Shared	M	LR	0.043	0.367	0.077	4234.2	16
Qwen3:8B	Shared	M	RF	0.080	0.375	0.132	2245.2	0
Qwen3:8B	Shared	M	gap	0.056	0.092	0.069	812.1	4
Qwen3:8B	Shared	T+M	LR	0.072	0.350	0.119	2349.4	21
Qwen3:8B	Shared	T+M	RF	0.043	0.400	0.077	4668.5	2
Qwen3:8B	Shared	T+M	gap	0.023	0.050	0.031	1129.1	23
Qwen3:8B	Joint	T	LR	0.022	0.300	0.041	6978.7	138
Qwen3:8B	Joint	T	RF	0.037	0.333	0.067	4555.7	82
Qwen3:8B	Joint	T	gap	0.034	0.025	0.029	373.8	16
Qwen3:8B	Joint	M	LR	0.121	0.375	0.183	1433.6	7
Qwen3:8B	Joint	M	RF	0.092	0.408	0.151	2115.2	0
Qwen3:8B	Joint	M	gap	0.049	0.017	0.025	171.5	4
Qwen3:8B	Joint	T+M	LR	0.070	0.375	0.117	2642.8	0
Qwen3:8B	Joint	T+M	RF	0.123	0.400	0.188	1508.3	3
Qwen3:8B	Joint	T+M	gap	0.015	0.033	0.021	1156.5	23
Qwen3:8B	Dedicated	none	all-active	0.028	0.033	0.030	621.3	24
Qwen3:8B	Dedicated	none	training-presence-prior	0.028	0.017	0.021	310.6	12
Qwen3:8B	Full	none	all-active	0.000	0.000	0.000	605.2	24
Qwen3:8B	Full	none	training-presence-prior	0.000	0.000	0.000	302.6	12
Qwen3:8B	Projected	none	all-active	0.000	0.000	0.000	625.4	24
Qwen3:8B	Projected	none	training-presence-prior	0.000	0.000	0.000	312.7	12
Qwen3:8B	Shared	none	all-active	0.035	0.042	0.038	603.6	24
Qwen3:8B	Shared	none	training-presence-prior	0.056	0.033	0.042	295.3	12
Qwen3:8B	Joint	none	all-active	0.035	0.042	0.038	611.2	24
Qwen3:8B	Joint	none	training-presence-prior	0.056	0.033	0.042	299.0	12

Table S37 continued

Executor	Condition	View	Detector	P	R	F1	FP/hour	Empty FP
Llama3.1:8B	Dedicated	T	LR	0.086	0.267	0.130	1105.9	61
Llama3.1:8B	Dedicated	T	RF	0.061	0.425	0.106	2569.6	83
Llama3.1:8B	Dedicated	T	gap	0.070	0.025	0.037	130.1	9
Llama3.1:8B	Dedicated	M	LR	0.194	0.358	0.251	582.2	2
Llama3.1:8B	Dedicated	M	RF	0.140	0.450	0.213	1079.9	0
Llama3.1:8B	Dedicated	M	gap	0.143	0.050	0.074	117.1	4
Llama3.1:8B	Dedicated	T+M	LR	0.169	0.425	0.242	813.2	1
Llama3.1:8B	Dedicated	T+M	RF	0.144	0.400	0.212	927.0	0
Llama3.1:8B	Dedicated	T+M	gap	0.055	0.050	0.052	338.3	16
Llama3.1:8B	Full	T	LR	0.027	0.158	0.046	2080.6	60
Llama3.1:8B	Full	T	RF	0.008	0.142	0.015	6593.2	56
Llama3.1:8B	Full	T	gap	0.000	0.000	0.000	2083.7	33
Llama3.1:8B	Full	M	LR	0.082	0.650	0.145	2676.4	0
Llama3.1:8B	Full	M	RF	0.035	0.492	0.066	4906.7	0
Llama3.1:8B	Full	M	gap	0.000	0.000	0.000	128.3	4
Llama3.1:8B	Full	T+M	LR	0.073	0.633	0.130	2969.7	0
Llama3.1:8B	Full	T+M	RF	0.037	0.492	0.069	4680.6	1
Llama3.1:8B	Full	T+M	gap	0.013	0.008	0.010	226.1	16
Llama3.1:8B	Projected	T	LR	0.022	0.142	0.038	2316.9	67
Llama3.1:8B	Projected	T	RF	0.007	0.125	0.013	6718.4	51
Llama3.1:8B	Projected	T	gap	0.000	0.000	0.000	2084.6	33
Llama3.1:8B	Projected	M	LR	0.040	0.142	0.062	1245.9	0
Llama3.1:8B	Projected	M	RF	0.009	0.167	0.017	6600.8	0
Llama3.1:8B	Projected	M	gap	0.000	0.000	0.000	1695.4	4
Llama3.1:8B	Projected	T+M	LR	0.037	0.242	0.065	2250.5	0
Llama3.1:8B	Projected	T+M	RF	0.023	0.350	0.044	5285.4	0
Llama3.1:8B	Projected	T+M	gap	0.000	0.000	0.000	2081.6	33
Llama3.1:8B	Shared	T	LR	0.114	0.350	0.171	1088.6	63
Llama3.1:8B	Shared	T	RF	0.059	0.433	0.103	2774.5	91
Llama3.1:8B	Shared	T	gap	0.071	0.083	0.077	431.4	16
Llama3.1:8B	Shared	M	LR	0.145	0.392	0.212	919.3	20
Llama3.1:8B	Shared	M	RF	0.174	0.508	0.259	962.5	0
Llama3.1:8B	Shared	M	gap	0.102	0.042	0.059	146.0	4
Llama3.1:8B	Shared	T+M	LR	0.144	0.433	0.216	1025.5	17
Llama3.1:8B	Shared	T+M	RF	0.186	0.375	0.249	653.8	0
Llama3.1:8B	Shared	T+M	gap	0.059	0.017	0.026	106.2	9
Llama3.1:8B	Joint	T	LR	0.107	0.250	0.150	795.8	47
Llama3.1:8B	Joint	T	RF	0.066	0.483	0.117	2594.2	143
Llama3.1:8B	Joint	T	gap	0.079	0.058	0.067	261.0	16
Llama3.1:8B	Joint	M	LR	0.182	0.358	0.242	614.3	4
Llama3.1:8B	Joint	M	RF	0.175	0.467	0.255	840.3	0
Llama3.1:8B	Joint	M	gap	0.205	0.075	0.110	111.4	4
Llama3.1:8B	Joint	T+M	LR	0.186	0.383	0.251	639.8	2
Llama3.1:8B	Joint	T+M	RF	0.199	0.442	0.274	681.2	2
Llama3.1:8B	Joint	T+M	gap	0.129	0.075	0.095	194.2	16
Llama3.1:8B	Dedicated	none	all-active	0.083	0.100	0.091	429.3	24
Llama3.1:8B	Dedicated	none	training-presence-prior	0.069	0.042	0.052	217.9	12
Llama3.1:8B	Full	none	all-active	0.000	0.000	0.000	440.0	24
Llama3.1:8B	Full	none	training-presence-prior	0.000	0.000	0.000	220.0	12
Llama3.1:8B	Projected	none	all-active	0.000	0.000	0.000	434.4	24
Llama3.1:8B	Projected	none	training-presence-prior	0.000	0.000	0.000	217.2	12
Llama3.1:8B	Shared	none	all-active	0.090	0.108	0.098	434.8	24
Llama3.1:8B	Shared	none	training-presence-prior	0.069	0.042	0.052	222.4	12
Llama3.1:8B	Joint	none	all-active	0.111	0.133	0.121	407.4	24
Llama3.1:8B	Joint	none	training-presence-prior	0.111	0.067	0.083	203.7	12

Table S37 continued

Executor	Condition	View	Detector	P	R	F1	FP/hour	Empty FP
Hosted alias	Full	T	LR	0.023	0.292	0.042	2193.0	21
Hosted alias	Full	T	RF	0.001	0.021	0.001	4895.3	28
Hosted alias	Full	T	gap	0.000	0.000	0.000	751.5	12
Hosted alias	Full	M	LR	0.024	0.458	0.047	3164.8	1
Hosted alias	Full	M	RF	0.008	0.208	0.015	4497.9	0
Hosted alias	Full	M	gap	0.008	0.042	0.013	928.5	1
Hosted alias	Full	T+M	LR	0.016	0.375	0.031	3974.1	1
Hosted alias	Full	T+M	RF	0.011	0.292	0.021	4519.6	0
Hosted alias	Full	T+M	gap	0.009	0.062	0.016	1156.1	13
Hosted alias	Projected	T	LR	0.057	0.479	0.101	1406.8	12
Hosted alias	Projected	T	RF	0.000	0.000	0.000	0.0	0
Hosted alias	Projected	T	gap	0.000	0.000	0.000	1792.5	13
Hosted alias	Projected	M	LR	0.027	0.354	0.050	2244.3	0
Hosted alias	Projected	M	RF	0.002	0.062	0.004	4969.9	0
Hosted alias	Projected	M	gap	0.062	0.042	0.050	110.2	1
Hosted alias	Projected	T+M	LR	0.038	0.521	0.071	2321.5	0
Hosted alias	Projected	T+M	RF	0.004	0.104	0.007	4712.7	1
Hosted alias	Projected	T+M	gap	0.019	0.042	0.026	374.7	13
Hosted alias	Full	none	all-active	0.009	0.021	0.013	386.6	12
Hosted alias	Full	none	training-presence-prior	0.014	0.021	0.017	256.5	8
Hosted alias	Projected	none	all-active	0.000	0.000	0.000	396.7	12
Hosted alias	Projected	none	training-presence-prior	0.000	0.000	0.000	264.5	8

References

[1] Y. Zhang, X. Deng, Z. Gu, Y. Chen, K. Xu, Q. Li, and J. Wu, "Exposing LLM user privacy via traffic fingerprint analysis: A study of privacy risks in LLM agent interactions," *arXiv preprint arXiv:2510.07176*, 2025. [Online]. Available: https://arxiv.org/abs/2510.07176v1

[2] H. Jeong, M. Teymoorianfard, A. Kumar, A. Houmansadr, and E. Bagdasarian, "Network-level prompt and trait leakage in local research agents," in *35th USENIX Security Symposium*. USENIX Association, 2026, pp. 6485–6504. [Online]. Available: https://www.usenix.org/conference/usenixsecurity26/presentation/jeong

[3] B. Dangol, "From privacy to workflow integrity: Communication-graph metadata in autonomous agent interoperability," *arXiv preprint arXiv:2606.07150v3*, 2026. [Online]. Available: https://arxiv.org/abs/2606.07150v3

[4] S. Pouryousef, J. Lopez, S. R. Nowmi, M. M. Kamol, M. Hossain, M. Tran, and M. S. Rahman, "A large-scale benchmark and risk assessment of traffic analysis attacks on cloud LLM services," *arXiv preprint arXiv:2609.31877v2*, 2026. [Online]. Available: https://arxiv.org/abs/2609.31877v2

[5] C. Song, H. Wang, and R. Martin, "PersonaFingerprint: Measuring persona inference on modern websites with LLM-driven browsing," *arXiv preprint arXiv:2605.15962v1*, 2026. [Online]. Available: https://arxiv.org/abs/2605.15962v1

[6] D. Kang, H. Jeong, J. Sheffey, P. Datta, and A. Houmansadr, "Whose agent are you? multi-layer fingerprinting and attribution of autonomous web agents," *arXiv preprint arXiv:2606.20910v1*, 2026. [Online]. Available: https://arxiv.org/abs/2606.20910v1

[7] I. Fayolle, S. Bouhenniche, S. Pélissier, P. Laperdrix, C. Maurice, and W. Rudametkin, "On the internet, nobody knows you're an LLM bot: Unmasking web agents with multi-layer fingerprinting," *arXiv preprint arXiv:2606.30119v1*, 2026. [Online]. Available: https://arxiv.org/abs/2606.30119v1

[8] A. Shahriar, M. N. Rahman, S. Ahmed, F. Sadeque, and M. R. Parvez, "AgentTell: Behavioural side-channel leakage in browser-use agents," *arXiv preprint arXiv:2609.32915v1*, 2026. [Online]. Available: https://arxiv.org/abs/2609.32915v1

[9] F. Li, X. Gao, Z. Liu, T. Li, C. Fu, Z. Wang, and K. Xu, "ANT: A multi-granularity network traffic dataset and benchmark for agents behavior auditing," *arXiv preprint arXiv:2610.06514v1*, 2026. [Online]. Available: https://arxiv.org/abs/2610.06514v1

[10] D. Ayzenshteyn, R. Weiss, and Y. Mirsky, "Cloak, honey, trap: Proactive defenses against LLM agents," in *34th USENIX Security Symposium*. USENIX Association, 2025, pp. 8095–8114. [Online]. Available: https://www.usenix.org/conference/usenixsecurity25/presentation/ayzenshteyn

[11] L. Song, Z. Pang, W. Wang, Z. Wang, X. Wang, H. Chen, W. Song, Y. Jin, D. Meng, and R. Hou, "The early bird catches the leak: Unveiling timing side channels in LLM serving systems," *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 11 431–11 446, 2025, cited version: arXiv:2409.20002v5. [Online]. Available: https://ieeexplore.ieee.org/document/11206380/

[12] G. McDonald and J. B. Or, "Whisper leak: a side-channel attack on large language models," *arXiv preprint arXiv:2511.03675*, 2025, cited version: arXiv:2511.03675v1. [Online]. Available: https://arxiv.org/abs/2511.03675v1

[13] T. Zhang, G. Saileshwar, and D. Lie, "Time will tell: Timing side channels via output token count in large language models," *arXiv preprint arXiv:2412.15431*, 2024, cited version: arXiv:2412.15431v1. [Online]. Available: https://arxiv.org/abs/2412.15431v1

[14] G. Wu, Z. Li, Y. Zhang, Z. Zhang, J. Niu, Y. Wu, and Y. Zhang, "Cacheprune: Privacy-aware and fine-grained KV cache sharing for efficient LLM inference," *arXiv preprint arXiv:2605.23640v1*, 2026. [Online]. Available: https://arxiv.org/html/2605.23640v1

[15] M. Juarez, S. Afroz, G. Acar, C. Diaz, and R. Greenstadt, "A critical evaluation of website fingerprinting attacks," in *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*, 2014, pp. 263–274.

[16] N. Mathews, J. K. Holland, S. E. Oh, M. S. Rahman, N. Hopper, and M. Wright, "SoK: A critical evaluation of efficient website fingerprinting defenses," in *IEEE Symposium on Security and Privacy*, 2023.

[17] T. Wang and I. Goldberg, "Walkie-talkie: An efficient defense against passive website fingerprinting attacks," in *26th USENIX Security Symposium*, 2017. [Online]. Available: https://www.usenix.org/conference/usenixsecurity17/technical-sessions/presentation/wang-tao

[18] J. Gong and T. Wang, "Zero-delay lightweight defenses against website fingerprinting," in *29th USENIX Security Symposium*. USENIX Association, 2020, pp. 717–734. [Online]. Available: https://www.usenix.org/conference/usenixsecurity20/presentation/gong

[19] F. Tramèr, N. Carlini, W. Brendel, and A. Madry, "On adaptive attacks to adversarial example defenses," in *Advances in Neural Information Processing Systems*, vol. 33, 2020. [Online]. Available: https://arxiv.org/abs/2002.08347v2

[20] R. Weiss, D. Ayzenshteyn, G. Amit, and Y. Mirsky, "What was your prompt? a remote keylogging attack on ai assistants," in *33rd USENIX Security Symposium*, 2024. [Online]. Available: https://www.usenix.org/conference/usenixsecurity24/presentation/weiss

[21] N. Carlini and M. Nasr, "Remote timing attacks on efficient language model inference," *arXiv preprint arXiv:2410.17175v1*, 2024. [Online]. Available: https://arxiv.org/html/2410.17175v1

[22] D. Ahmed, A. Sabir, and A. Das, "Spying through your voice assistants: Realistic voice command fingerprinting," in *32nd USENIX Security Symposium*, 2023, pp. 2419–2436. [Online]. Available: https://www.usenix.org/conference/usenixsecurity23/presentation/ahmed-dilawer

[23] Model Context Protocol, "Transports," 2025, protocol version 2025-06-18. [Online]. Available: https://modelcontextprotocol.io/specification/2025-06-18/basic/transports

[24] M. Thomson and C. Benfield, "HTTP/2," RFC 9113, 2022. [Online]. Available: https://www.rfc-editor.org/rfc/rfc9113.html

[25] E. Rescorla, "The transport layer security (tls) protocol version 1.3," RFC 8446, 2018. [Online]. Available: https://www.rfc-editor.org/rfc/rfc8446.html

[26] W. Lugoloobi, S. Marro, J. Magomere, J. Wright, and C. Russell, "Known by their actions: Fingerprinting LLM browser agents via UI traces," *arXiv preprint arXiv:2605.14786v1*, 2026. [Online]. Available: https://arxiv.org/abs/2605.14786v1

[27] J. Roh, E. Bagdasarian, H. Haddadi, and A. S. Shamsabadi, "SPILLage: Agentic oversharing on the web," *arXiv preprint arXiv:2602.13516v1*, 2026. [Online]. Available: https://arxiv.org/abs/2602.13516v1

[28] S. Chen, R. Wang, X. Wang, and K. Zhang, "Side-channel leaks in web applications: A reality today, a challenge tomorrow," in *IEEE Symposium on Security and Privacy*, 2010. [Online]. Available: https://staff.ie.cuhk.edu.hk/~khzhang/my-papers/2010-oakland-side-channel.pdf

[29] P. Chapman and D. Evans, "Automated black-box detection of side-channel vulnerabilities in web applications," in *ACM Conference on Computer and Communications Security*, 2011. [Online]. Available: https://www.cs.virginia.edu/~evans/pubs/ccs2011/

[30] P. Reynolds, J. L. Wiener, J. C. Mogul, M. K. Aguilera, and A. Vahdat, "WAP5: Black-box performance debugging for wide-area systems," in *International World Wide Web Conference*, 2006. [Online]. Available: https://cseweb.ucsd.edu/~vahdat/papers/wap5-www06.pdf

[31] B. C. Tak, C. Tang, C. Zhang, S. Govindan, B. Urgaonkar, and R. N. Chang, "vPath: Precise discovery of request processing paths from black-box observations of thread and network activities," in *USENIX Annual Technical Conference*, 2009. [Online]. Available: https://www.usenix.org/conference/usenix-09/vpath-precise-discovery-request-processing-paths-black-box-observations-thread

[32] K. P. Dyer, S. E. Coull, T. Ristenpart, and T. Shrimpton, "Peek-a-boo, i still see you: Why efficient traffic analysis countermeasures fail," in *IEEE Symposium on Security and Privacy*, 2012. [Online]. Available: https://pages.cs.wisc.edu/~rist/papers/trafanal.pdf

[33] R. Trimananda, J. Varmarken, A. Markopoulou, and B. Demsky, "Packet-level signatures for smart home devices," in *Network and Distributed System Security Symposium*, 2020. [Online]. Available: https://www.ndss-symposium.org/ndss-paper/packet-level-signatures-for-smart-home-devices/

[34] N. Apthorpe, D. Y. Huang, D. Reisman, A. Narayanan, and N. Feamster, "Keeping the smart home private with smart(er) IoT traffic shaping," *Proceedings on Privacy Enhancing Technologies*, vol. 2019, no. 3, pp. 128–148, 2019. [Online]. Available: https://petsymposium.org/popets/2019/popets-2019-0040.pdf

[35] D. Duan, Q. Liu, Y. Liu, N. Zou, X. Hu, and D. Liu, "How much can tool calls reveal about you? privacy leakage from tool-call traces," *arXiv preprint arXiv:2610.41885v1*, 2026. [Online]. Available: https://arxiv.org/abs/2610.41885v1

[36] Y. Yao, Z. Wang, H. Cheng, Y. Cheng, H. Du, and X.-Y. Li, "IntentMiner: Intent inversion attack via tool call analysis in the model context protocol," *arXiv preprint arXiv:2512.14166v2*, 2026. [Online]. Available: https://arxiv.org/abs/2512.14166v2

[37] P. Sirinam, M. Imani, M. Juarez, and M. Wright, "Deep fingerprinting: Undermining website fingerprinting defenses with deep learning," in *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*, 2018. [Online]. Available: https://arxiv.org/abs/1801.02265v5

[38] S. Bhat, D. Lu, A. Kwon, and S. Devadas, "Var-CNN: A data-efficient website fingerprinting attack based on deep learning," *Proceedings on Privacy Enhancing Technologies*, vol. 2019, no. 4, pp. 292–310, 2019. [Online]. Available: https://petsymposium.org/popets/2019/popets-2019-0070.php

[39] M. Bishop, "HTTP/3," RFC 9114, 2022. [Online]. Available: https://www.rfc-editor.org/rfc/rfc9114.html

[40] J. Iyengar and M. Thomson, "QUIC: A UDP-based multiplexed and secure transport," RFC 9000, 2021. [Online]. Available: https://www.rfc-editor.org/rfc/rfc9000.html

[41] C. Song, D. D. M. Mekala, H. Wang, and R. P. Martin, "Redefining website fingerprinting attacks with multi-agent LLMs," *Proceedings on Privacy Enhancing Technologies*, vol. 2026, no. 3, pp. 688–702, 2026. [Online]. Available: https://www.petsymposium.org/popets/2026/popets-2026-0101.php

[42] S. Zavrak, "Content-aware attack detection in LLM agent tool-call traffic: An empirical study of features, architectures, and evaluation protocols," *arXiv preprint arXiv:2605.11053v3*, 2026. [Online]. Available: https://arxiv.org/abs/2605.11053v3

[43] Z. Guan, G. Xiong, G. Gou, Z. Li, M. Cui, and C. Liu, "BAPM: Block attention profiling model for multi-tab website fingerprinting attacks on Tor," in *Annual Computer Security Applications Conference*, 2021.

[44] X. Deng, Q. Yin, Z. Liu, X. Zhao, Q. Li, M. Xu, K. Xu, and J. Wu, "Robust multi-tab website fingerprinting attacks in the wild," in *IEEE Symposium on Security and Privacy*, 2023, pp. 1005–1022.

[45] T. Wang and I. Goldberg, "On realistically attacking Tor with website fingerprinting," *Proceedings on Privacy Enhancing Technologies*, vol. 2016, no. 4, pp. 21–36, 2016.

[46] J. Hayes and G. Danezis, "k-fingerprinting: A robust scalable website fingerprinting technique," in *25th USENIX Security Symposium*, 2016, pp. 1187–1203. [Online]. Available: https://www.usenix.org/conference/usenixsecurity16/technical-sessions/presentation/hayes

[47] G. Zhao, H. Wu, and X. Zhang, "AgentTrace: Execution-level fingerprinting of black-box LLM agents," 2026, version 1. [Online]. Available: https://arxiv.org/abs/2610.32092v1

[48] M. Ediga and S. Chattopadhyay, "Trace: Unmasking AI attack agents through terminal behavior fingerprinting," 2026, version 1. [Online]. Available: https://arxiv.org/abs/2605.01186v1

[49] O. Borysenko, "Developer experience with AI coding agents: HTTP behavioral signatures in documentation portals," 2026, extended version 3. [Online]. Available: https://arxiv.org/abs/2604.02544v3

[50] ——, "Documentation retrieval by AI coding agents: An empirical study of HTTP behavioral signatures," in *Proceedings of the 1st International Workshop on Agentic AI for Next-Generation Software Development*. ACM, 2026, pp. 168–169.

[51] Reworr and D. Volkov, "LLM agent honeypot: Monitoring AI hacking agents in the wild," 2025, version 2; first submitted 2024. [Online]. Available: https://arxiv.org/abs/2410.13919v2

[52] D. Arp, E. Quiring, F. Pendlebury, A. Warnecke, F. Pierazzi, C. Wressnegger, L. Cavallaro, and K. Rieck, "Dos and don'ts of machine learning in computer security," in *31st USENIX Security Symposium (USENIX Security 22)*. USENIX Association, 2022, pp. 3971–3988. [Online]. Available: https://www.usenix.org/conference/usenixsecurity22/presentation/arp

[53] Y. Yao, J. Duan, K. Xu, Y. Cai, Z. Sun, and Y. Zhang, "A survey on large language model (LLM) security and privacy: The good, the bad, and the ugly," *arXiv preprint arXiv:2312.02003v3*, 2024. [Online]. Available: https://arxiv.org/abs/2312.02003v3

[54] A. Shahriar, M. N. Rahman, S. Ahmed, F. Sadeque, and M. R. Parvez, "A survey on agentic security: Applications, threats and defenses," *arXiv preprint arXiv:2510.06445v1*, 2025. [Online]. Available: https://arxiv.org/abs/2510.06445v1

[55] M. A. Ferrag, N. Tihanyi, D. Hamouda, L. Maglaras, and M. Debbah, "From prompt injections to protocol exploits: Threats in LLM-powered AI agents workflows," *arXiv preprint arXiv:2506.23260v1*, 2025. [Online]. Available: https://arxiv.org/abs/2506.23260v1

[56] LangChain, "LangGraph, version 1.0.3," Software release, November 10, 2025. [Online]. Available: https://pypi.org/project/langgraph/1.0.3/

[57] E. Debenedetti, I. Shumailov, T. Fan, J. Hayes, N. Carlini, D. Fabian, C. Kern, C. Shi, A. Terzis, and F. Tramèr, "Defeating prompt injections by design," 2025, version 2. [Online]. Available: https://arxiv.org/abs/2503.18813v2

[58] M. Costa, B. Köpf, A. Kolluri, A. Paverd, M. Russinovich, A. Salem, S. Tople, L. Wutschitz, and S. Zanella-Béguelin, "Securing AI agents with information-flow control," 2025, version 2. [Online]. Available: https://arxiv.org/abs/2505.23643v2

[59] T. Shi, J. He, Z. Wang, H. Li, L. Wu, W. Guo, and D. Song, "Progent: Programmable privilege control for LLM agents," 2025, version 2. [Online]. Available: https://arxiv.org/abs/2504.11703v2

[60] S. Yao, N. Shinn, P. Razavi, and K. Narasimhan, "τ-bench: A benchmark for tool-agent-user interaction in real-world domains," 2024, version 1. [Online]. Available: https://arxiv.org/abs/2406.12045v1

[61] E. Debenedetti, J. Zhang, M. Balunović, L. Beurer-Kellner, M. Fischer, and F. Tramèr, "AgentDojo: A dynamic environment to evaluate prompt injection attacks and defenses for LLM agents," in *Advances in Neural Information Processing Systems*, vol. 37, 2024. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2024/hash/97091a5177d8dc64b1da8bf3e1f6fb54-Abstract-Datasets_and_Benchmarks_Track.html

[62] X. Cai, R. Nithyanand, T. Wang, R. Johnson, and I. Goldberg, "A systematic approach to developing and evaluating website fingerprinting defenses," in *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security*, 2014, pp. 227–238.